PROGRESS REPORT II

项目进度报告（二）

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ABSTRACT

Elevators are equipped in almost every modern building, and millions of people take elevators every day. However, series of elevator accidents have been reported in recent years, and Chinese society has paid unprecedented attention to elevator ride safety as well as quality. In the first phase of this project we have established both vertical and horizontal vibration model to study the vibration characteristics of elevator system and analyzed the behaviors under free vibration as well as forced vibration conditions. Further in phase 2, lab experimental as well as in–situ test were carried out and specific analysis methods & indicators are proposed for elevator system fault diagnosis, while corresponding main achievements can be summarized as:

1. Measurement scenario for pilot elevator system;
2. Condition monitoring and fault diagnosis method for traction motor, guide rail/shoe and hoist rope;
3. 3 selected time domain indicators, 3 new feature frequency indicators, 2 integrated indicators and an integrated energy ratio histogram are proposed for diagnosing 3 types of rope fault;
4. A self–checking scenario and fault database for elevator monitoring and diagnosis.

The rest of this report is structured as follows. Firstly, we introduce the significance and objects of vibration–based monitoring and diagnosis for elevator system in Section 1. In Section 2, the research background is presented. In Section 3, we summarize the scenario and content of elevator vibration measurement on pilot elevator as well as rope tester. In Section 4, we put emphasis on traction motor fault diagnosis and show our research outcome. Guide rail/shoe fault diagnosis, i.e., horizontal ride quality evaluation is studied in Section 5. Then in Section 6, we elaborate the research on hoist rope and proposed condition monitoring indicators. In Section 7, all former proposed methods and indicators are integrated into a self–checking scenario and fault database for real world application. Finally the conclusion and plans for the future work are suggested in Section 8.

Key words: Elevator vibration, dynamics model, solution, frequency analysis

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1 OBJECTS

Elevators are of rapidly increasing significance along with the quick city development process. For elevator passengers, the safety as well as ride quality are of key importance. Vibration–based analysis methods are widely used in machinery fault diagnosis, owing to its high accuracy and efficiency. However, related research especially research on specific diagnosis method and indicators have been limited, not to mention the application in general elevators.

Aiming at ensuring the safety and comfort during elevator running, the objective of the proposed project is to deepen the understanding of available vibration and force measurements thus providing reliable guidance for the ride quality monitoring and fault diagnosis of elevator system. Both parties agree the proposed work will focus on the measurement and analysis of rope force as well as car vibration signals, thus revealing the running condition of the objective elevator and obtaining effective fault diagnosis methods as well as indicators.

The project encompasses three main phases, while phase 2 focus on the vibration measurement and diagnosis method for elevator system, especially the motor, horizontal guide rail and most of all, the ropes. Main tasks of this project phase include:

Task 1: Scenario for measurement of key physical quantities, including hoist rope tension, car vibration in three dimensions and background noise. (Car displacement, velocity and change rate of acceleration can be obtained by integration or differential of the vertical vibration.)

Task 2: Experimental evaluation of the former proposed vibration model, particularly designing experiments for collecting elevator signals under different stages, loads and failure conditions like pulley wear, rope force imbalance, counterweight imbalance, etc. Data acquisition under carefully designed experimental settings will be carried out on the objective elevator by Tsinghua with supported from KONE.

Task 3: Preprocessing of the collected data in order for convenience and better accuracy in further analysis in stage 3. Signal transform approaches like envelope extraction, noise suppression, etc., are applied when necessary. The merits of these preprocessing will also be demonstrated via comparison.

2 RESEARCH BACKGROUND

The evolution of equipment maintenance mode has experienced three main phases. In 1950s the equipment are maintained in breakdown maintenance (BM) mode, while equipment are repaired only after breakdown. It caused huge property and life loss. Then in 1960s, preventive maintenance (PM) mode was proposed to plan for regular maintenance. The drawback of preventive maintenance, maintenance resources waste, had been recognized in 1980s and condition based maintenance (CBM) were proposed. Thereafter, measurement and fault diagnosis techniques have been rapidly developing. Nowadays a derivative of condition based maintenance termed reliability centered maintenance (RCM) is popular all around the world, which focus on the readability of equipment and proactive maintenance. For elevator system, the condition based maintenance and reliability centered maintenance are both developing and condition monitoring and fault diagnosis for elevator are still under hot research.

Elevator measurement techniques are developing rapidly in these years. Acceleration characteristics is a key index for elevator ride quality and acceleration measurement has evolved from the early WR7300F recorder by GRAPHETEC Inc to micro and digital measurement system like DT-2 elevator acceleration tester [1]. Compound vibration measurement system like ADIAS and EVA-625 have also been wide applied in car vibration and noise measurement. They can further recognize the acceleration, velocity, displacement and further carry out simple analysis. Besides, vibration measurement for traction motor has also been under study and a 911KT traction motor vibration tester has been developed and used in motor fault diagnosis [2]. Guide rail relates much to the system horizontal vibration and Tianjin University developed a guide rail detecting system for guide rail measurement [3]. Elevator balance coefficient is one of the parameters which ensure safety, and balance coefficient measurement based on displacement transducer, current and voltage transducer have been developed [4].

Besides, the elevator diagnosis methods have also been in development owing to more powerful measurement equipment and algorithms. Special fault detection techniques like ultrasonic flaw detection and laser collimator [5] have been applied in elevator fault diagnosis field. Nevertheless, the research achievements on elevator machinery fault diagnosis have been very limited. Since the theory of vibration based

analysis are widely studied, the elevator fault diagnosis method mostly fall in this field. A few work on elevator fault diagnosis technique have been reported [6-11]. However, specific fault diagnosis method or indicators are very limited [12-15], and most diagnosis method doesn't have innovative breakthrough in fault characteristics and inherent indicators.

Intelligent & networked monitoring and control system has been applied to elevators in most big elevator companies [16-18]. It helps record elevator health information remotely and quick react to accident. However, current monitoring systems mostly focus on basic running parameters like acceleration, velocity and noise level, while mechanical characteristics which are related to fault evolution still demand for deep research.

3 VIBRATION MEASUREMENT

A KONE 3000X MiniSpace elevator (36594114) is chosen as the project pilot elevator. The mechanical structure and main configuration parameters have been elaborated in project report 1. In this phase, project team study the measurement points for vibration test and managed to establish a test scenario with in–situ application. Specific test on rope fault characteristics were also carried out on a lab rope tester to simulate different types of rope fault. In this section, the detail of vibration measurement scenario and in–situ application are exhibited.

3.1 Vibration test on pilot elevator

Mechanical vibration signals contain rich abundant of machinery health information. By measuring and analyzing elevator vibration signals from different components, we are able to study its dynamic characteristics. With regard to test on pilot elevator, the test scenario is summarized as follows:

3.1.1 Measuring points

(1) Traction motor:

As the driving part and power source, traction motor is a main component which demands vibration monitoring. Besides, for the pilot elevator its traction motor is fixed on roof. Hence, we employed a set of wired vibration data acquisition system and 7 accelerometers in total were mounted on the motor case, as presented in Figure 3.1. A laser speed tachometer was also used to monitor motor running speed.

Fig 3.1 Motor vibration measuring points

(2) Guide rail:

Horizontal vibration suppression contributes much to the elevator ride quality. Since the car is vertically in motion, we mounted the accelerometer horizontally on fixed guide rail. 4 wireless accelerometers were evenly distributed on the guide rail, as shown in Figure 3.2

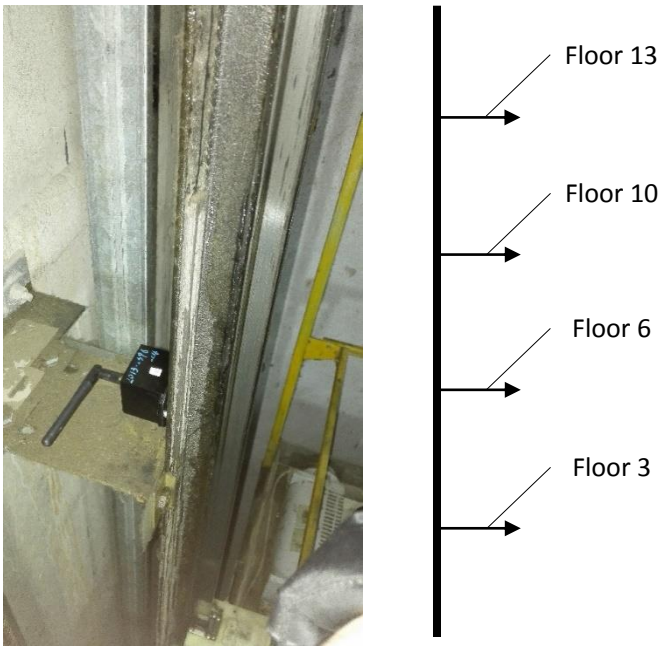


Fig 3.2 Guide rail vibration measuring points

(3) Car:

The car vibration is of extreme importance since it is where passenger lies. Three unidirectional accelerometers were mounted on the car casing, as given in Figure 3.3.

Fig 3.3 Car vibration measuring points

(4) Car frame:

Damping layer is set between car and car frame in order for car vibration suppression. Therefore, valuable mechanical vibration information are filtered out in car vibration signal. We measured the car frame vibration with 3 unidirectional accelerometers, as can be seen in Figure 3.4.

Fig 3.4 Car frame vibration measuring points

(5) Counterweight:

For comparison with car frame vibration, counterweight vibration in vertical direction was also measured using a wireless accelerometer, see Figure 3.5.

Fig 3.5 Counterweight vibration measuring points

(6) Pulleys:

Since the hoist rope go through the whole elevator system and is rich in vibration information, we measured the pulley vibration to indirectly monitor rope condition. 2 wireless accelerometers were mounted respectively in radial and axial direction of car frame pulley, as exhibited in Figure 3.6.

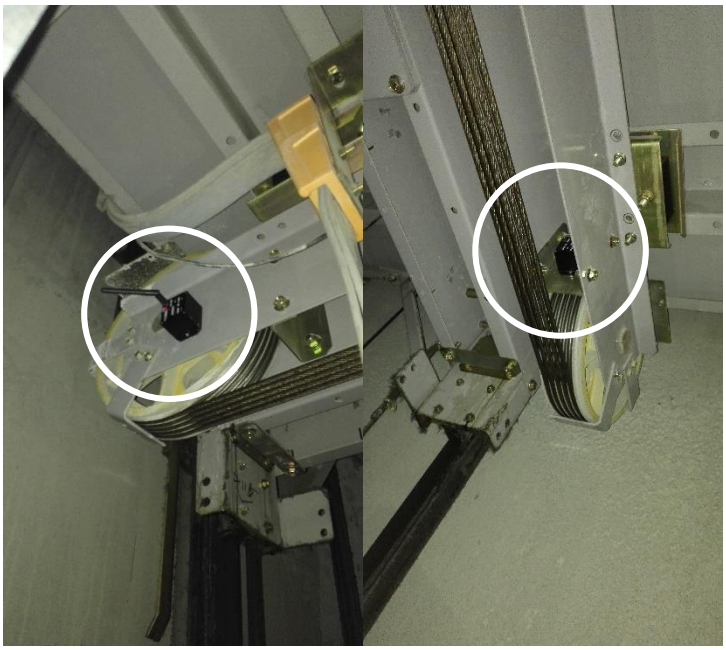


Fig 3.6 Pulley vibration measuring points

(7) Overall distribution:

To summarize, the overall measuring points distribution is illustrated in Figure 3.7.

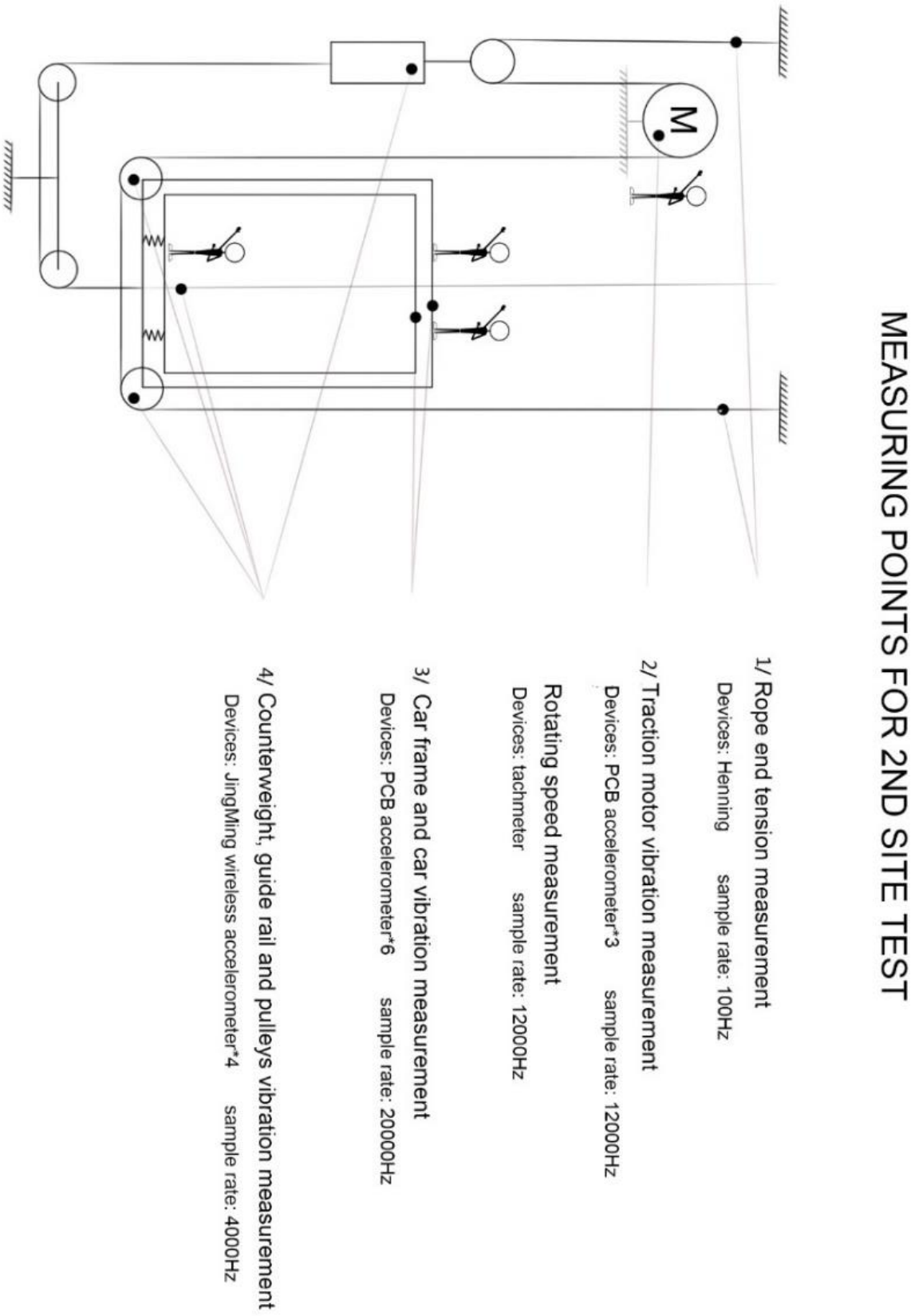


Fig 3.7 Overall vibration measuring points

3.1.2 Test scenario

(1) Vertical modal test:

Natural vibration contribute much to the elevator system vibration. In order to study the natural frequencies of elevator system and their varying trend, we carried out a series of modal test.

Initial excitation is the key to arise free natural vibration. For vertical modal test, since the common force hammer is small in size and unable to provide enough initial force, we came up with a new way. The car travel from 3-13 floor and braked suddenly at each floor height. By measuring elevator vibration with aforementioned sensors, we are able to analyze the frequency constitution of free vibration signal and thus identifying natural frequencies. Besides, test under 2 different load conditions were also carried out. The scenario of modal test on car is also recorded in Table 3.1.

Table 3.1 Vertical modal test scenario

No.	**Location**	**Load (car frame)**	**Load (car)**	**Sampling time**	**Note**	**File**
1	Floor 13	3*60kg	2*60kg	10s	Brake	Xx01
2	Floor 12					
3	Floor 11					
……	……	……	……	……	……	……
9	Floor 5					
10	Floor 4					
11	Floor 3					
12	Floor 13	3*60kg	2*60kg+100kg	10s	Brake	Xx12
13	Floor 12					
15	Floor 11					
……	……	……	……	……	……	……
20	Floor 5					
21	Floor 4					
22	Floor 3					

(2) Car travel test:

Actually the car runs between floors smoothly in everyday service. We also carried out travel test to examine the dynamic characteristics. With the same transducer distribution, scenario of car travel test is given in Table 3.2.

Table 3.2 Car modal test scenario

No.	Location	Load (car frame)	Load (car)	Sampling time	Note	File
23	Floor 13~floor 3	3*60kg	2*60kg	40s	Smooth	Xx23
24	Floor 3~floor 8					
25	Floor 8~floor 13					
26	Floor 13~floor 8					
27	Floor 8~floor 3					
28	Floor 3~floor 13					
29	Floor 13~floor 3				4 wireless accelerometers on guide rail	
30	Floor 3~floor 13				4 wireless accelerometers on guide rail	

(3) Horizontal rail modal test:

For horizontal modal test, we hammered on the car casing in both directions on horizontal plane and meanwhile collected vibration signals. Besides, the guide rail vibration also contributes to the system horizontal vibration. We also hammer on the guide rail on 4 different positions to test its natural characteristics. Scenario for horizontal modal test is exhibited in Table 3.3.

Table 3.3 Car modal test scenario

No.	Location	Load (car frame)	Load (car)	Sampling time	Note	File
31	Floor 13	3*60kg	2*60kg	40s	Hammer in x direction	Xx31
24	Floor 13				Hammer in y direction	
25	Floor 13				Hammer on guide rail	
26	Floor 8				Hammer in x direction	
27	Floor 8				Hammer in y direction	
28	Floor 8				Hammer on guide rail	
29	Floor 3				Hammer in x direction	

30	Floor 3				Hammer in y direction	
31	Floor 3				Hammer on guide rail	

3.2 Vibration test on rope tester

Project team also carried out special test on a rope tester in kunshan to further study the mechanical characteristics of hoist rope and simulate rope fault for evaluating fault diagnosis method.

3.2.1 Measuring points

(1) Drive motor:

Although the output end of driving motor is connected to a crank connecting rod for transforming rotating motion into reciprocating motion, we try to monitor its vibration for reference. The running speed of outer end sheave was also recorded by a speed tachometer. Picture of motor and connecting gearbox is given in Figure 3.8.

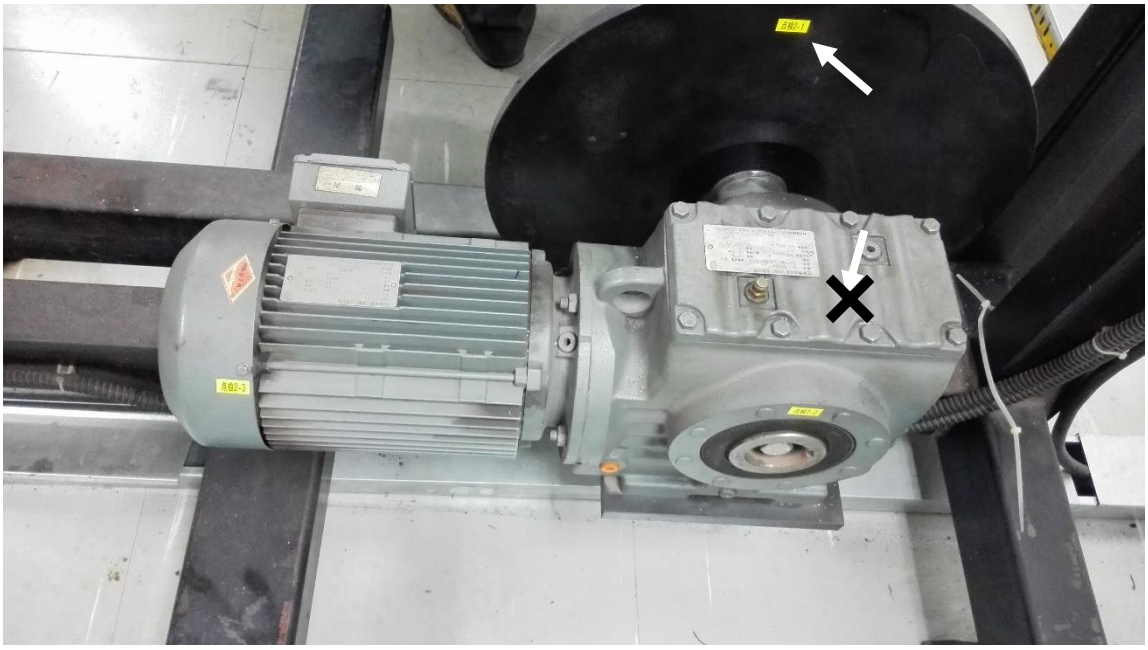


Fig 3.8 Driving motor vibration measuring points

(2) Pulley:

Three main pulleys exist in the rope tester for reciprocating fatigue test. In this test we focus on the method of monitoring the rope condition by pulley vibration measurement and analysis. Three accelerometers in different directions were mounted on the base of pulley shaft. In addition, a laser tachometer was also employed to measure the pulley linear velocity, as can be seen in Figure 3.9.

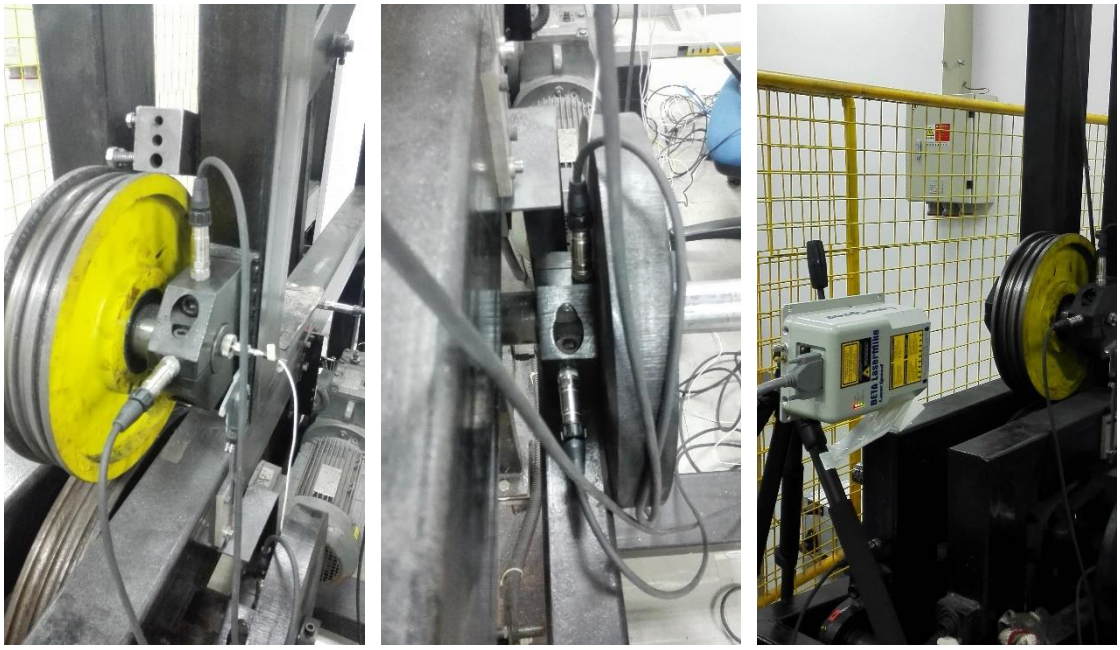


Fig 3.9 Rope tester pulley vibration measuring points

(3) Framework:

Except for these key components, in fact the framework of rope tester should also be put emphasis on. To study the natural vibration characteristics of the rope tester, 2 extra sensors were mounted on the rope tester framework in both directions on horizontal plane, as illustrated in Figure 3.10.

Fig 3.10 Rope tester framework vibration measuring points

(4) Overall distribution:

To summarize, the overall measuring points distribution is illustrated in Figure 3.11.

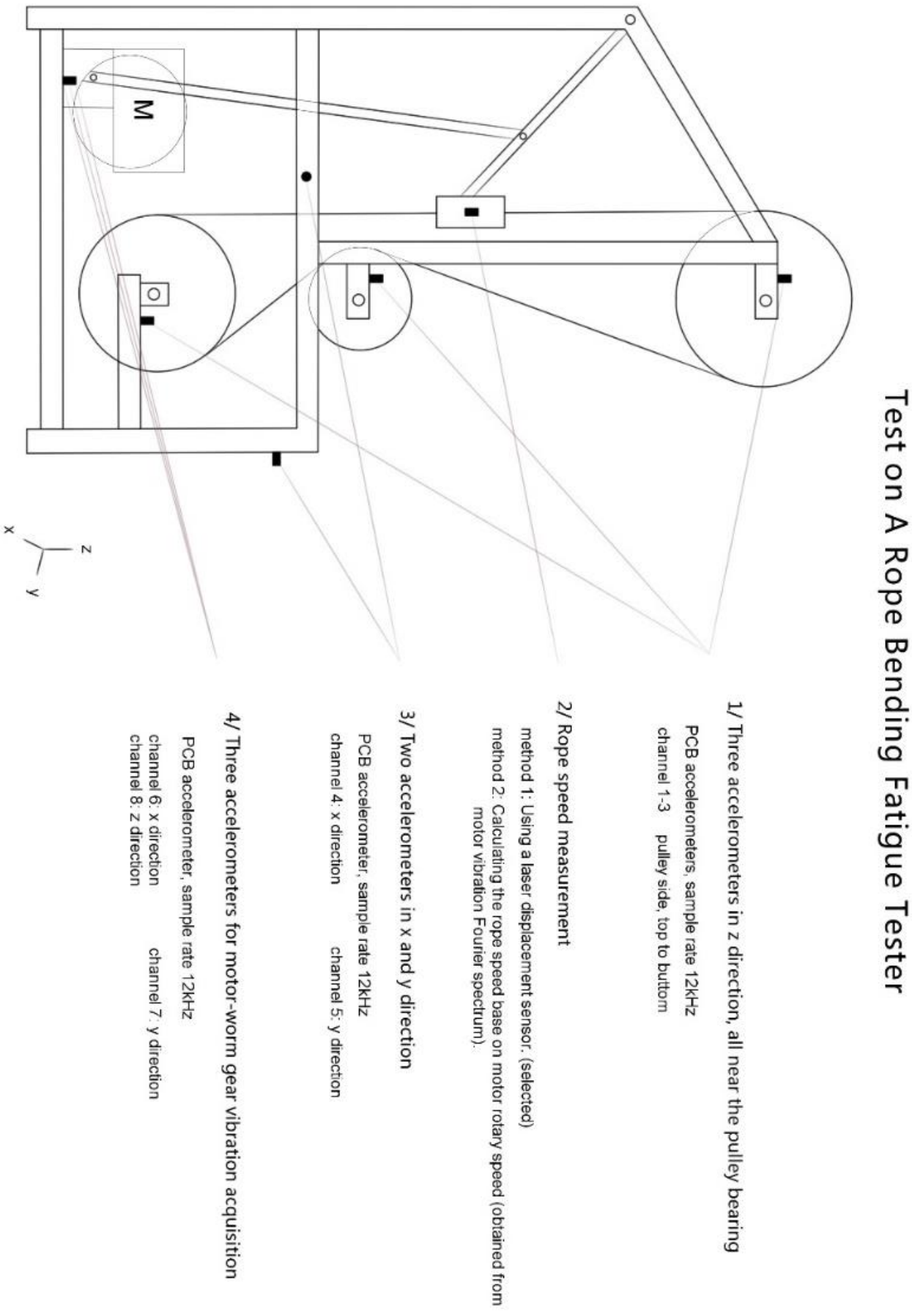


Fig 3.11 Overall vibration measuring points

3.2.2 Rope fault simulation

(1) Baseline:

Since both new and old (fatigued) 13mm ropes were available, we choose the 13mm rope as research object. Basic size information of this type of rope is shown in Figure 3.12.

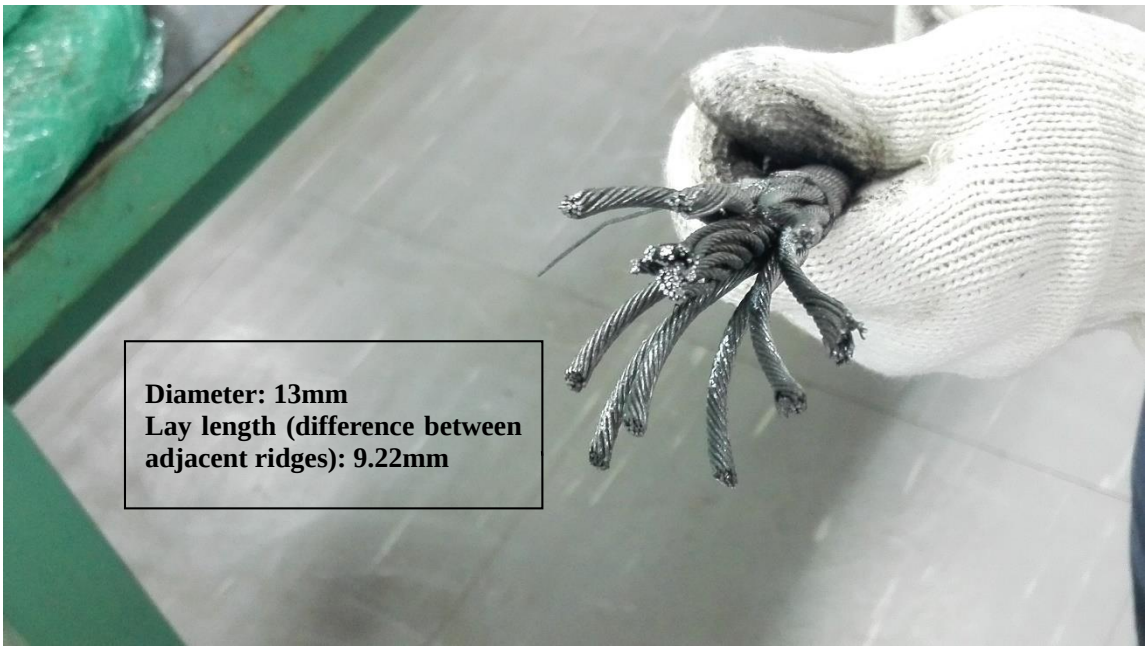


Fig 3.12 Baseline rope

(2) Rope fatigue:

After years' service, the hoist rope may become fatigued, which is a complex type of compound fault. The fatigued rope was provided by KONE after an accelerated test, as presented in Figure 3.13.

Fig 3.13 Fatigued rope

(3) Rope with local bump:

In order to simulate local fault, we added a circle of iron wire around a healthy rope, see Figure 3.14. It will be used to study rope local fault diagnosis.

Fig 3.14 Rope with local bump

(4) Rope with local breakage:

In addition, we introduce local breakage fault into a new rope to simulate local fault, as given in Figure 3.15.

Fig 3.15 Rope with local breakage

3.2.3 Test scenario

Test under 4 different rope conditions were carried out on the rope tester. Besides, we introduce different load condition as well. Modal test and regular running test were recorded in Table 3.4.

Table 3.4 Vertical modal test scenario

No.	Rope condition	Load (car frame)	Sampling time	Note	File
1	Fatigue	Full load	60s	Modal test	Xx01
2		Full load	60s * 3 times	Running	Xx02-04
3		Empty load	60s	Modal test	Xx05
4		Empty load	60s * 3 times	Running	Xx06-08
5	Baseline	Empty load	60s	Modal test	Xx09
6		Empty load	60s * 3 times	Running	Xx10-12
7		Full load	60s	Modal test	Xx13
8		Full load	60s * 3 times	Running	Xx14-16
9	Bump	Full load	60s * 3 times	Running	Xx17-19
10	Breakage	Full load	60s	Modal test	Xx20
11		Full load	60s * 3 times	Running	Xx21-23
12		Empty load	60s	Modal test	Xx24
13		Empty load	60s * 3 times	Running	Xx25-27

4 ROPE FAULT DIAGNOSIS

Hoist rope is the key component to connect each part as a whole system. It has different fault types and result in poor ride quality or even car dropping. However, monitoring the rope condition and diagnose rope fault is challenging due to its unique structure and dynamic characteristics. In this section we studied rope fault characteristics and propose fault monitoring indicators in both time and frequency domain.

4.1 Data acquisition

Since hoist rope is running and difficult for vibration measurement, **we propose to collect and analyze the pulley vibration signal which reflects much information of rope.** We applied specific test on rope under 3 types of fault based on lab experiment as well as numerical simulation.

4.2 Distributed fault diagnosis

A common rope fault type is surface wear, which makes the rope vulnerable and brings in extra vibration. To detect rope wear fault, we proposed a feature frequency indicator termed rope passing frequency.

4.2.1 Principle

The idea of rope passing frequency is motivated by the surface structure of rope. As illustrated in Figure 4.1, there are cyclic ridges and pits on the surface of rope. When the rope runs through a pulley, the contact stiffness varies at a frequency which can be cast as

$$f_{\mathrm{p}} = \frac{v}{l}, \tag{4.1}$$

where f_{p} stands for rope passing frequency, v is the rope linear velocity and l means the interval between adjacent ridges, i.e., lay length divide number of ropes.

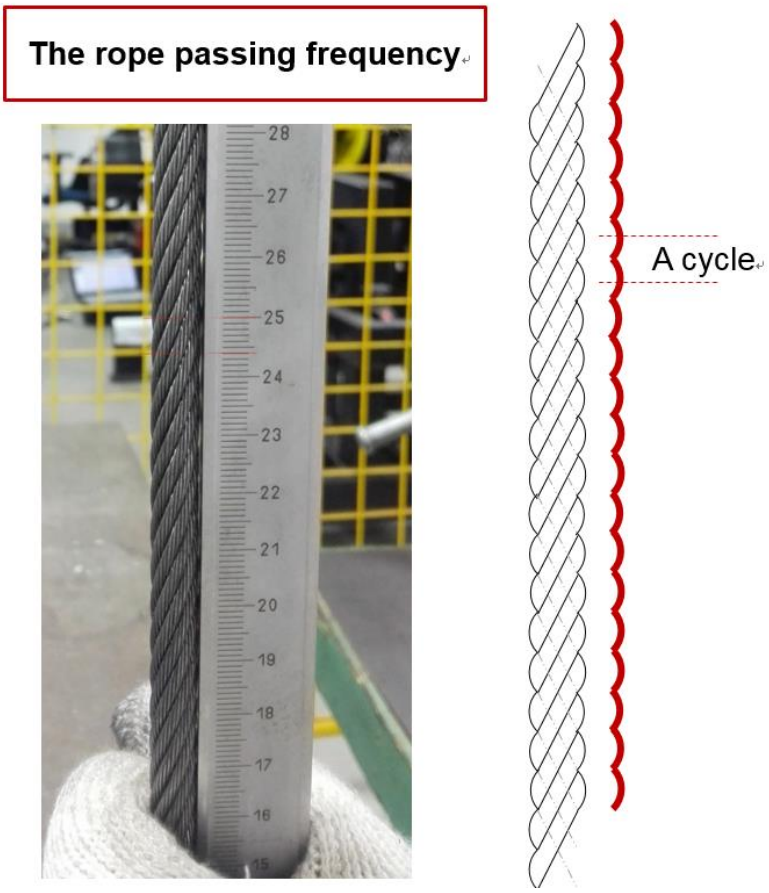


Fig 4.1 Rope surface structure

4.2.2 Validation

(1) Test on pilot elevator:

For an 8mm rope used in pilot elevator, its lay length is measured as 6mm, therefore in pilot elevator test, the theoretical rope passing frequency is calculated as

$$f_{\mathrm{p}} = \frac{v}{l} = \frac{3.19}{6*10^{-3}}\,\mathrm{Hz} = 531\mathrm{Hz}\,. \tag{4.2}$$

To validate the existence of rope passing frequency, we firstly perform a comparison between Fourier spectra of both pulley vibration signal as well as car frame vibration signal. The measuring points of both signals are given in Figure 4.2 and the Fourier spectra comparison is shown in Figure 4.3. Obviously a peak marked with black circle is observed. The peak is of high amplitude only in pulley vibration signal and is around 531Hz, thus it is validated as the rope passing frequency.

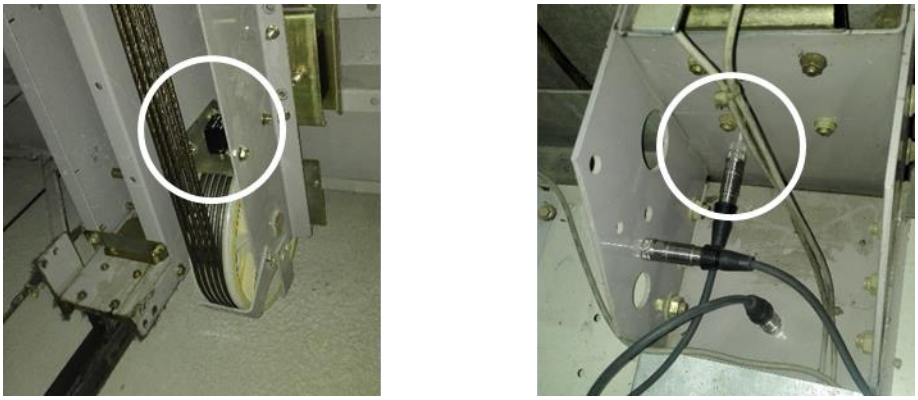


Fig 4.2 Measuring points

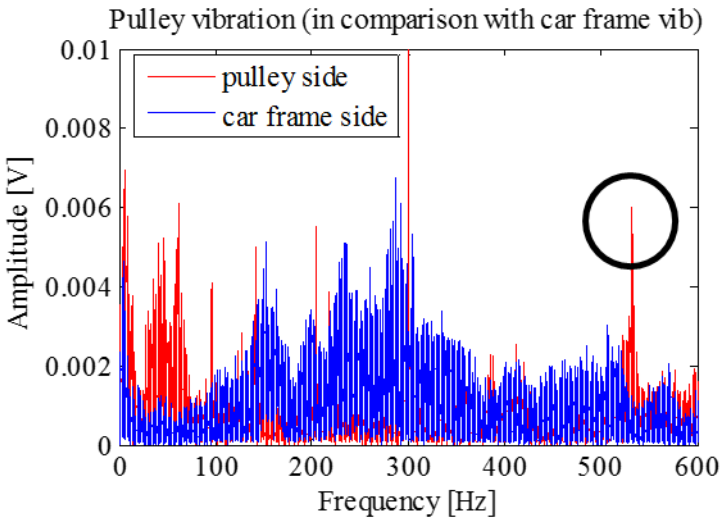


Fig 4.3 Fourier spectra comparison

In addition, we try to detect the rope passing frequency in Fourier spectrum of motor vibration signal. The theoretical rope passing frequency is the same as equation 4.2. As can be seen in Figure 4.4, the 531Hz frequency peak is also identified, thus validating the existence of rope passing frequency.

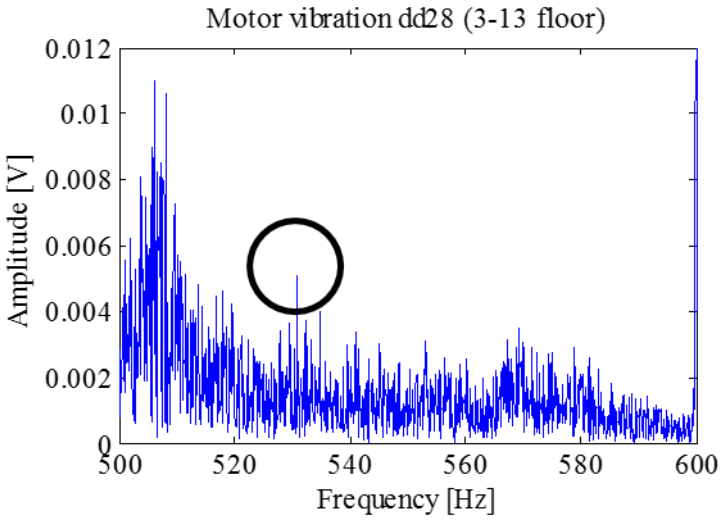


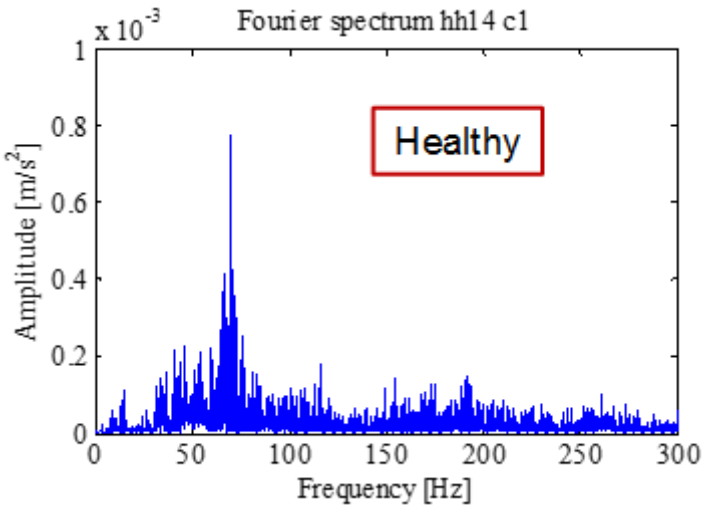
Fig 4.4 Fourier spectrum of motor vibration signal

(2) Test on rope tester:

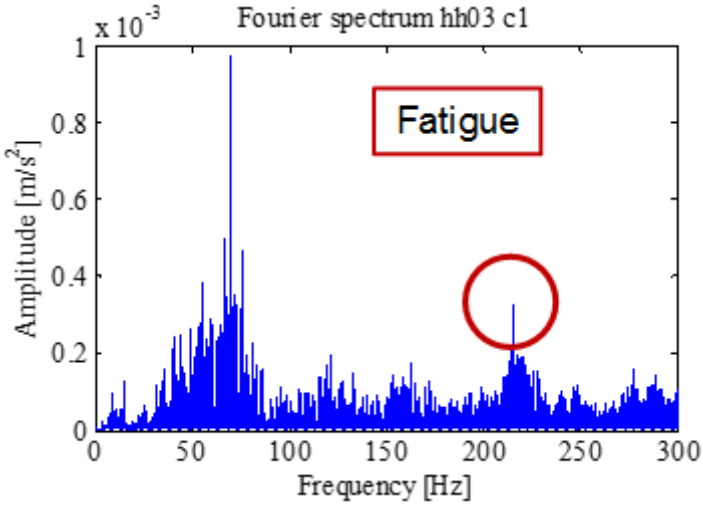
The rope used in rope tester has a diameter of 13mm, therefore the theoretical rope passing frequency is calculated as

$$f_{\mathrm{p}} = \frac{v}{l} = \frac{2.17}{9.22*10^{-3}}\,\mathrm{Hz} = 235.3\,\mathrm{Hz}\,. \tag{4.3}$$

It should be noted that the velocity here (2.17m/s) is the maximum rope linear speed. Considering the periodically time–varying linear velocity of ropes, the time–frequency analysis is expected. However, the rope tester suffers from severe impact at route ends which hinders the identification of characteristic frequency components. Fortunately the nonstationary feature in Fourier spectrum is not evident for limited range frequency fluctuation. Fourier spectra of pulley vibration signals under both healthy and rope fatigue conditions are displayed in Figure 4.5(a) and (b). By comparison we clearly observe an extra frequency peak in Figure 4.5(b), which is around 235Hz and validated as the rope passing frequency.



(a) Healthy condition



(b) Fatigue condition

Fig 4.4 Fourier spectrum of rope tester pulley vibration signal

4.2.3 Usage

Since the amplitude of rope passing frequency in fatigue case is much higher than that in healthy case, we use the rope passing frequency as a frequency domain indicator for rope fault diagnosis.

Former analysis imply that the amplitude of rope passing frequency may not be in proportion to rope fault severity. According to the rope–pulley structure we propose a rope degradation assumption while 5 phases are included in life time process. The degradation process and corresponding indicator varying trend is plotted in Figure 4.5. Different from conventional time domain indicators, the amplitude of rope passing frequency is supposed to increase during early stage and drop a little when bumps are worn out. In late phase it will increase since the vibration enhances. The procedure of monitoring rope health condition with rope passing frequency is summarized as follows:

(1) Collect pulley vibration signal under fixed working condition;
(2) Obtain the amplitude of rope passing frequency by Fourier spectrum analysis. The energy in the vicinity of rope passing frequency can also serve as the indicator;
(3) Record the value of indicator and get the monitoring chart;
(4) Take moves when the indicator has obvious varying trend.

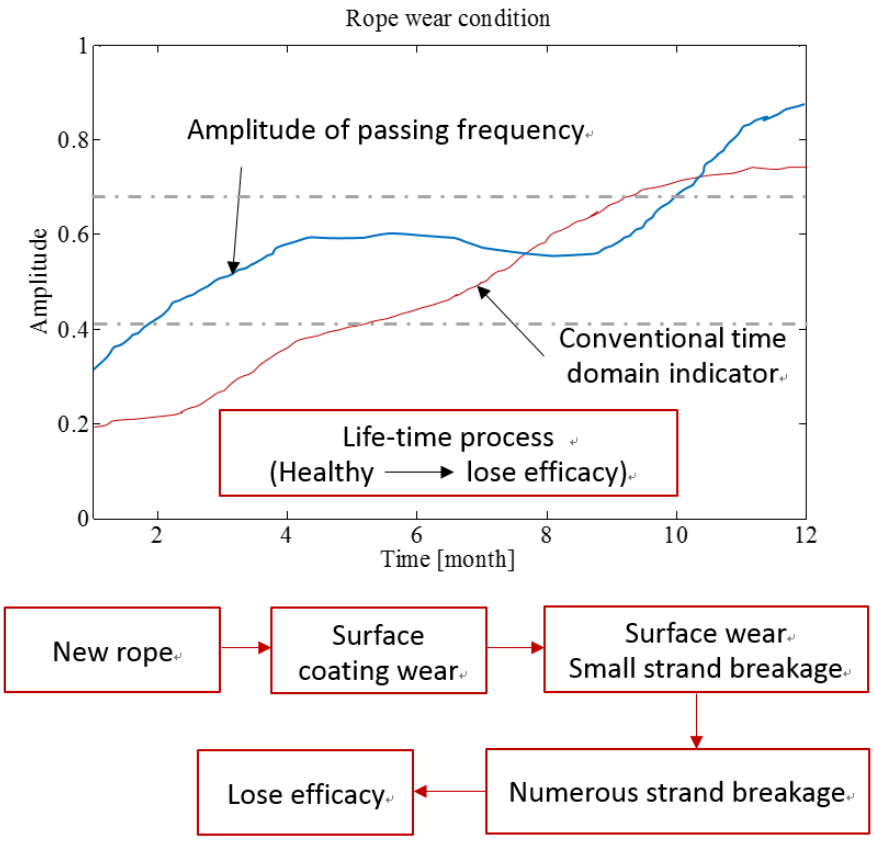


Fig 4.5 Degradation process

4.3 Rope connection fault diagnosis

Many of reported elevator accidents were due to rope loose or even derail. This category of rope connection fault has direct influence on system mechanical structure and result in the change of stiffness and other parameters. Hence, we study the characteristics of elevator system and propose the 3rd natural frequency as feature frequency indicator.

4.3.1 Principle and validation

According to former achievements in project phase 1, we have built a system vibration model to analyze its vibration properties, as can be seen in Figure 4.6.

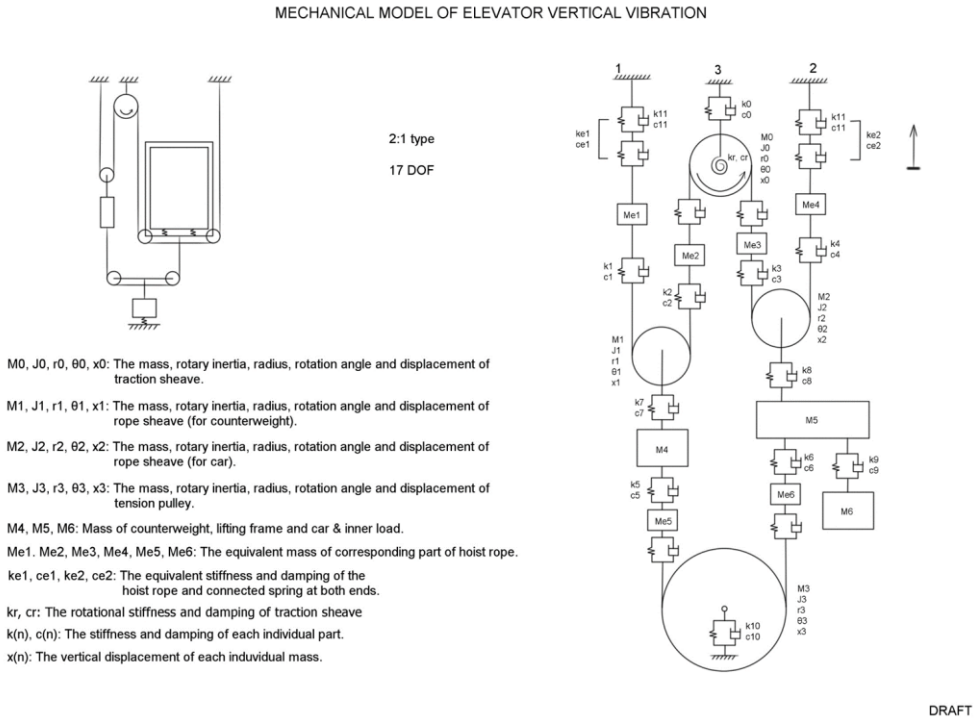


Fig 4.6 Vertical vibration model

Based on the mathematical model on Matlab platform, we carried out fault simulation to analyze the influence brought in with rope loose. As presented in Figure 4.7, 5 different rope conditions are simulated and the instantaneous value of 3rd natural frequency are plotted. The reason why we choose 3rd natural frequency as the indicator is that this order of natural frequency is most evident in Fourier spectrum of car frame vibration signals (see Figure 4.8), and is easy for extraction.

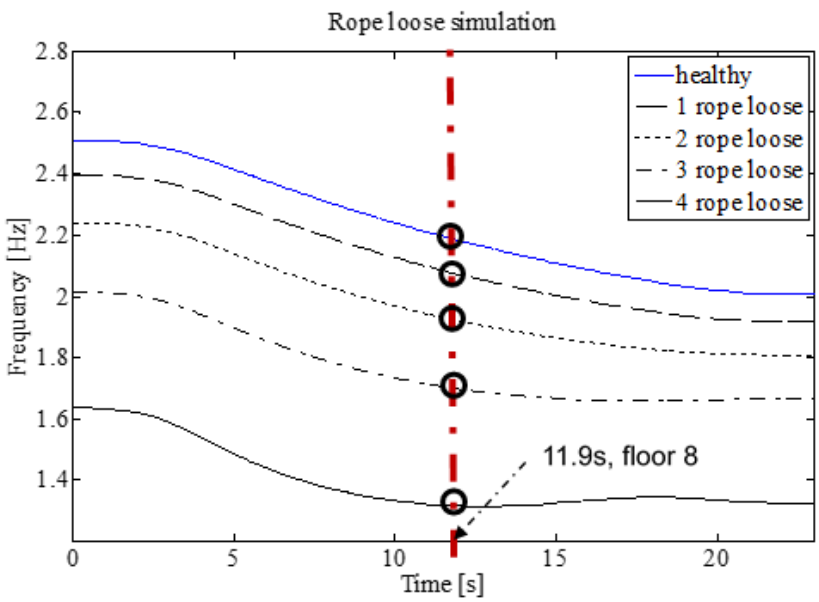


Fig 4.7 3rd natural frequency

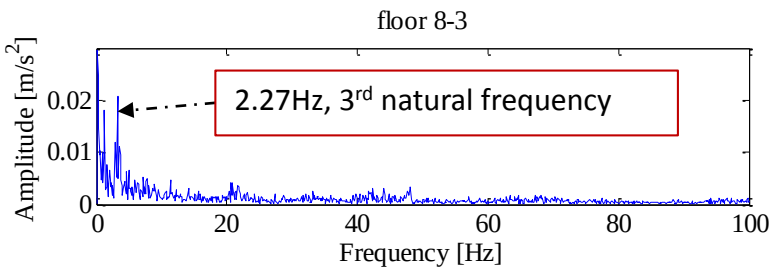


Fig 4.8 ourier spectrum of car frame vibration signal

According to Figure 4.7, the 3rd natural frequency decrease in value once the rope is loose. When more ropes become loose, the decreasing amplitude is even higher. Hence, the value of 3rd natural frequency can indicate rope connection information. We take the 3rd natural frequency as a feature frequency indicator for rope connection fault diagnosis.

4.3.2 Usage

(1) Fault monitoring:

Based on above description, we have found that the value of 3rd natural frequency decreases in proportional to rope loose severity. Hence, by monitoring the value of 3rd natural frequency in Fourier spectrum of car frame vibration signal, we are able to monitor the rope connection condition. The illustration is plotted in Figure 4.9.

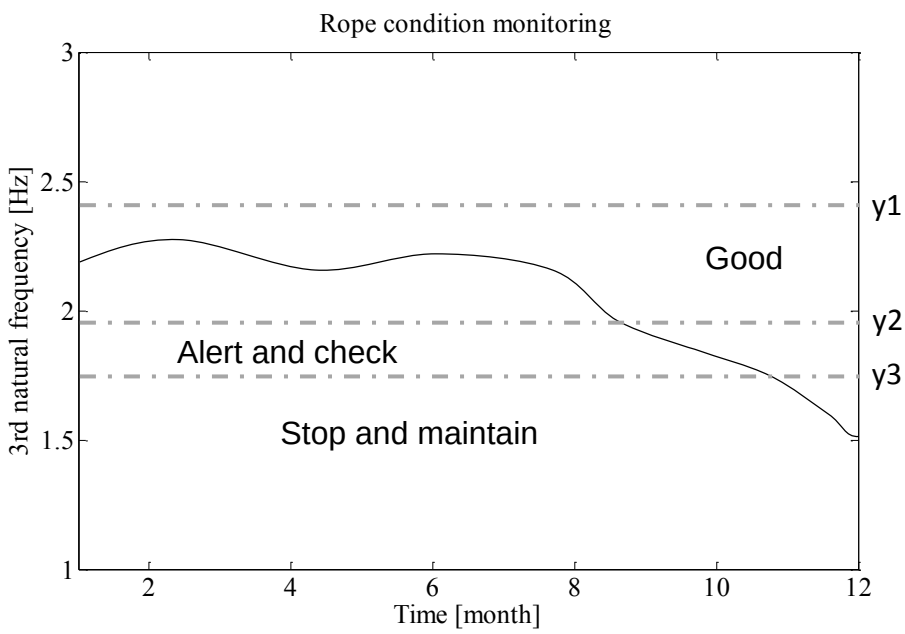


Fig 4.9 Monitoring of 3rd natural frequency

The procedure is summarized as:

(1) After the installation of an elevator, perform a modal test (8th floor) or journey test (3-13th floor) and get the value of dominant natural frequency;

(2) Monitor the value of dominant natural frequency termly;
(3) Record the value of indicator every constant times and get the monitoring chart;
(4) Take moves when the indicator has obvious varying trend.

(2) Fault localization:

Since the rope has two ends, we propose a method to further localize the fault thus saving maintenance cost. Still based on fault simulation when one of the rope end is loose while the other is normal, the variation of 3rd natural frequency is different, as shown in Figure 4.10.

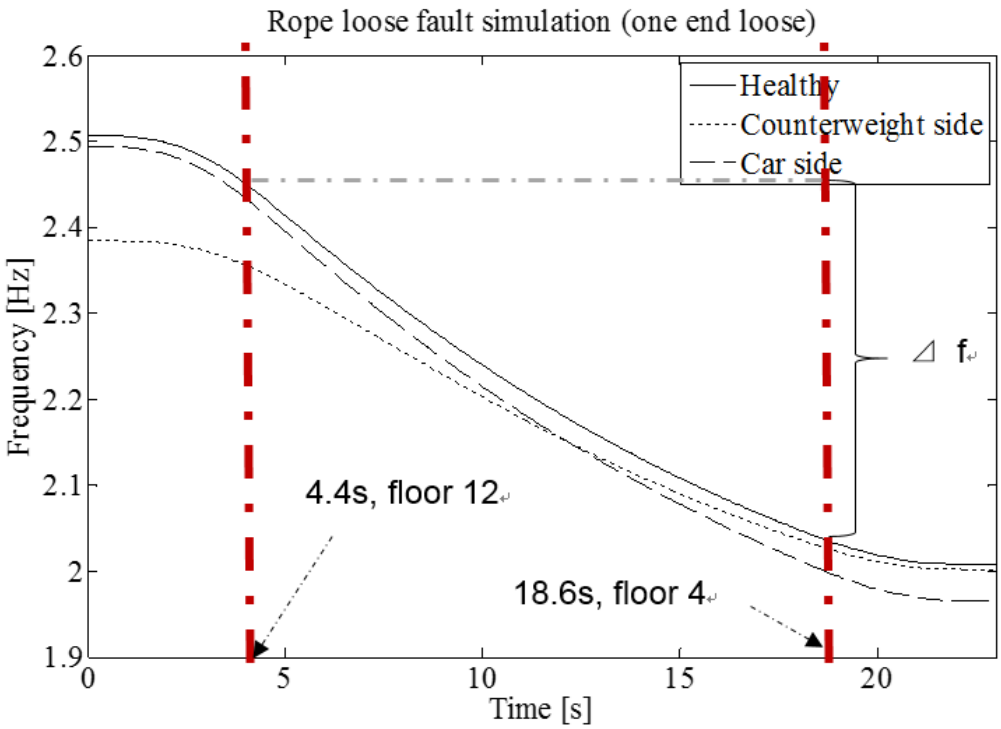


Fig 4.10 Localize rope fault with 3rd natural frequency

Two modal tests or two time periods of travel test are supposed to be carried out on both high floor (floor 12) and low floor (floor 4). The 3rd natural frequency obtained in these two cases are recorded as f_1 and f_2, while we take their difference Δf as the indicator. According to the result of rope loose fault simulation, Δf increases if the car side rope end is loose, while the decrease indicates counterweight side rope end loose. Usage of this indicator is further illustrated in Figure 4.11.

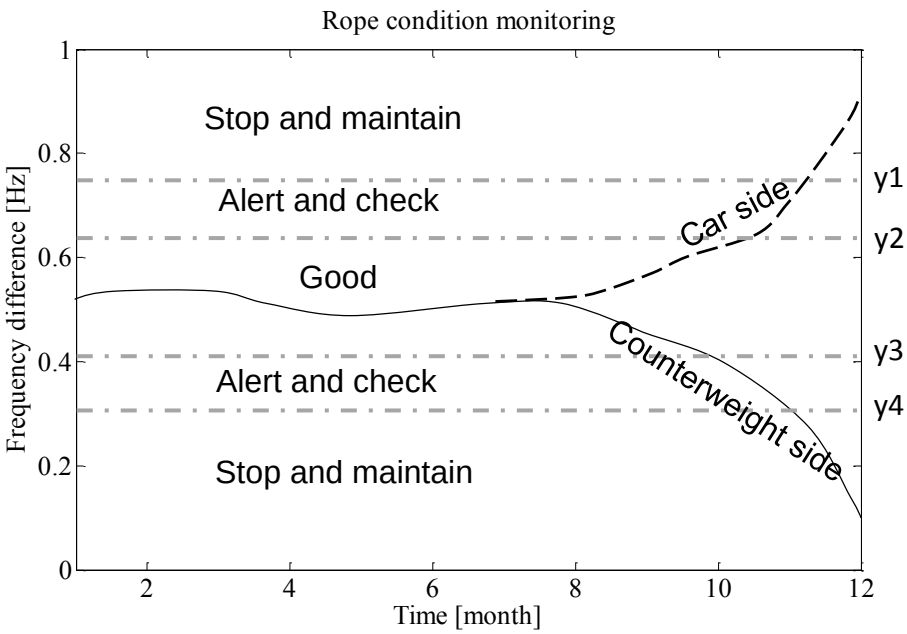


Fig 4.11 Usage of 3rd natural frequency difference

The procedure is summarized as:

(1) Calculate the difference between 3rd natural frequency at 4th floor and 12th floor, as Δ f;

(2) Monitor the value of Δ f, as illustrated above;

(3) Record the value of indicator every constant times and get the monitoring chart;

(4) Take moves when the indicator has obvious varying trend.

(3) Improvement:

Since it is prone to error pinpointing the value of the natural frequency directly from Fourier spectrum, especially under nonstationary conditions, a new indicator $E(f_1,f_2)$ standing for signal power within certain frequency range is proposed. The definition of $E(f_1,f_2)$ is given in equation 4.4, and illustrated in Figure 4.12. It can serve as an improved version of the former proposed indicator as value of 3rd natural frequency. When E decreases, more severe fault exists. Its usage can refer to Figure 4.9.

$$E(f_1, f_2) = \sum_{f=f_1}^{f_2} A_f^2 \ , \tag{4.4}$$

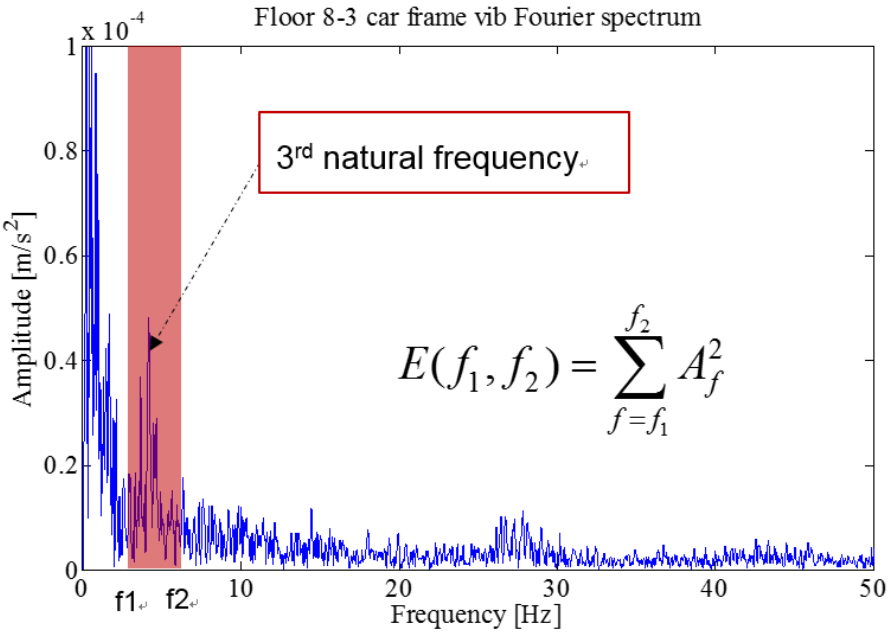


Fig 4.12 Definition of frequency range energy

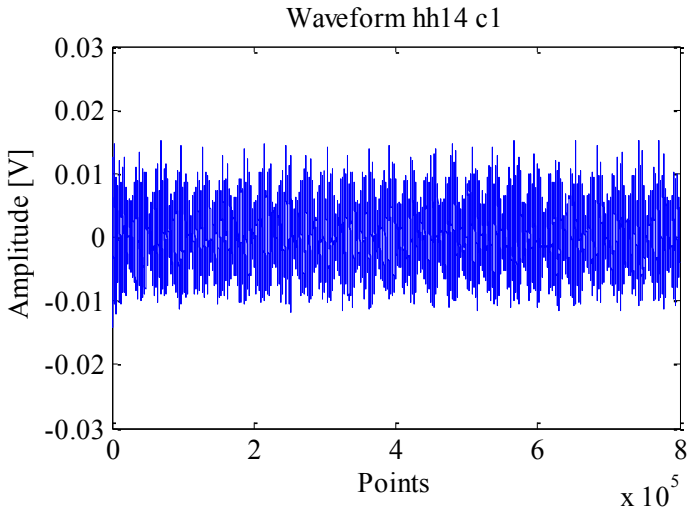
An energy ratio indicator is further proposed to help judge the fault location (car side or counterweight side). The definition is shown in equation 4.5. It can serve as an improved version of the former proposed indicator as difference between 3rd natural frequency on high floor as well as low floor. Its usage can refer to Figure 4.11.

$$ER = \frac{E_{\text{low}}(f_1, f_2)}{E_{\text{high}}(f_1, f_2)}, \tag{4.5}$$

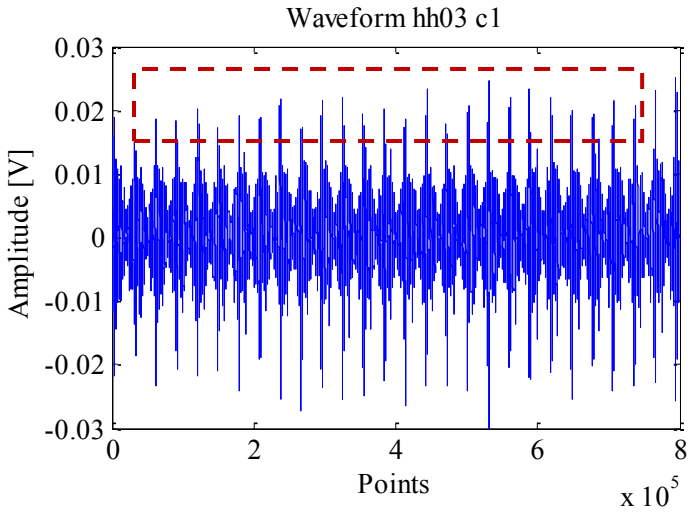
4.4 Time domain diagnosis method

4.4.1 Conventional waveform analysis

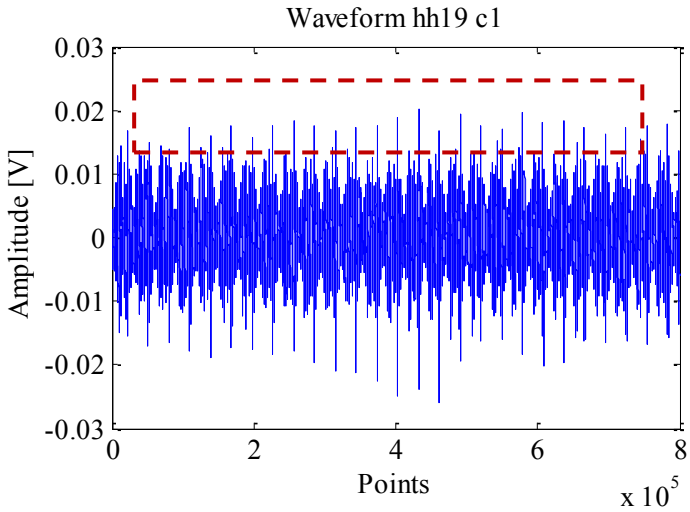
Waveform analysis is a simple tool for coarse fault diagnosis. In Figure 4.13 displayed the pulley vibration signal waveform under baseline, rope fatigue, rope local bump and rope local breakage cases.



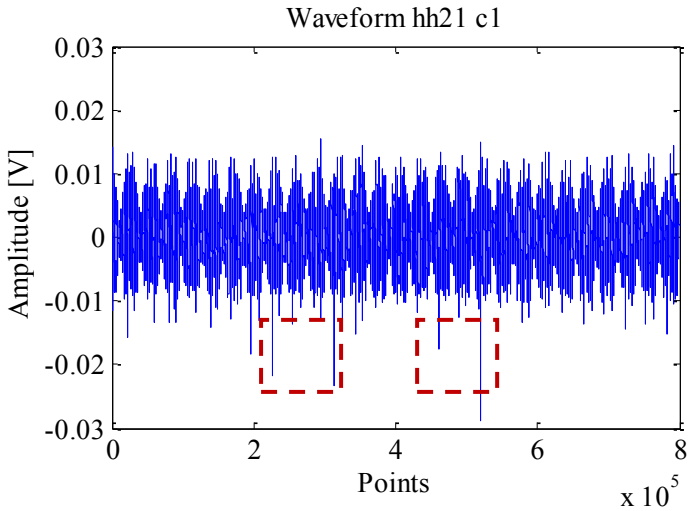
(a) Baseline



(b) Rope fatigue



(c) Rope local bump



(d) Rope local breakage

Fig 4.13 Waveform of pulley vibration signal

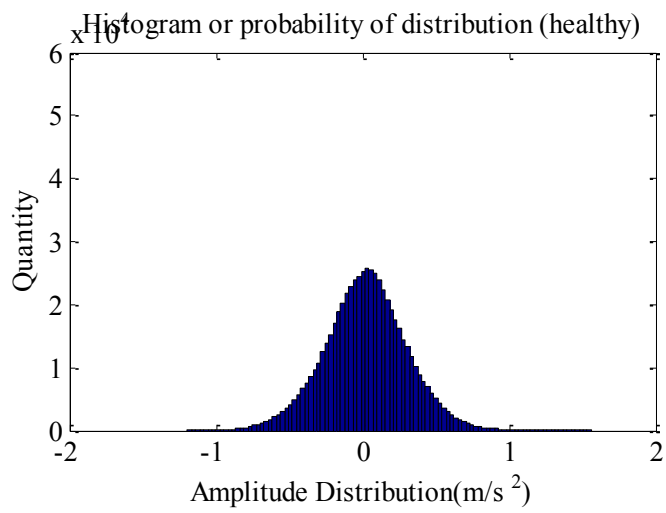
From waveform of pulley vibration signals, we observe evident impulses when fatigue fault is brought in. However, in case of local bump fault, although the difference between baseline and faulty one is still obvious, but impulses in waveform is weaker than fatigue case. What is worse, in case of local breakage fault, the difference in waveform is actually inconspicuous. Therefore, traditional waveform detection is not capable of diagnosing all fault types.

4.4.2 Probability density distribution

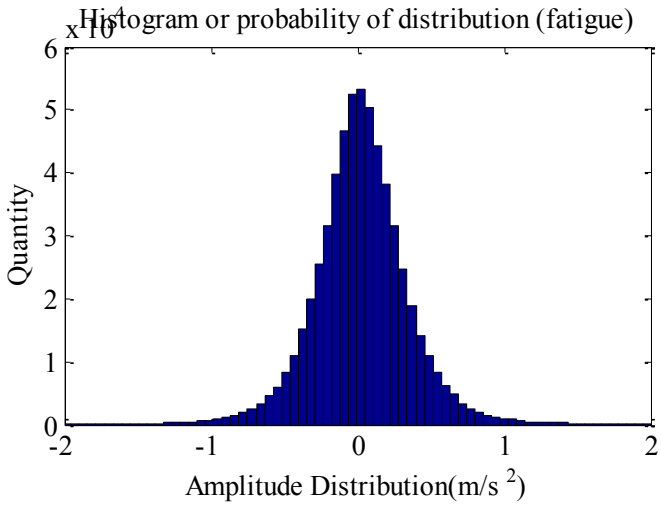
Probability density distribution is another time domain tool for coarse fault diagnosis. Its definition is cast in equation 4.6.

$$p(x) = \lim_{\Delta x \to 0} \frac{P[x \le x(t) \le x + \Delta x]}{\Delta x} = \lim_{\Delta x \to 0} \frac{1}{\Delta x} [\lim_{T \to \infty} \frac{\Delta t}{T}] . \tag{4.6}$$

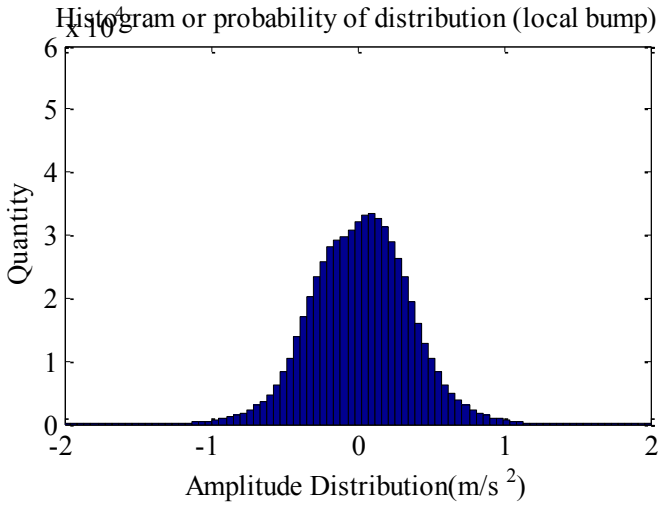
Histogram of probability density distribution under aforementioned 4 rope conditions are exhibited in Figure 4.14.



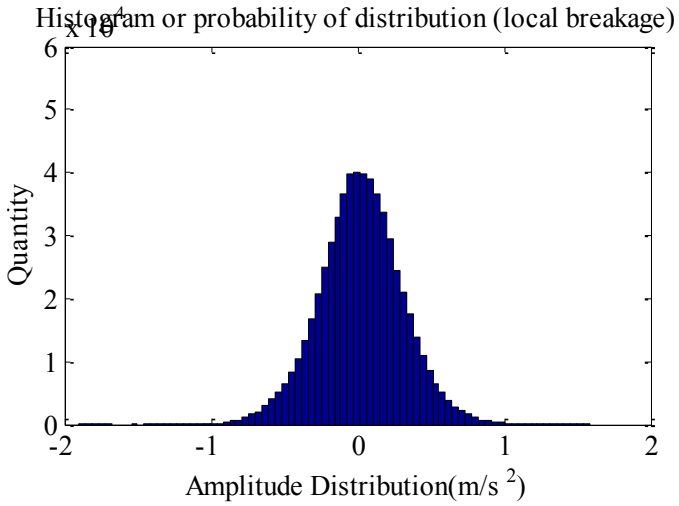
(a) Baseline



(b) Rope fatigue



(c) Rope local bump



(d) Rope local breakage

Fig 4.14 Waveform of pulley vibration signal

Probability density distribution, as illustrated above, is probably the simplest monitoring method for it does not involve complex calculation. However, it is not suitable for precise diagnosis. More feasible shape indicators (height, intersection area, etc) can also be drawn from this distribution.

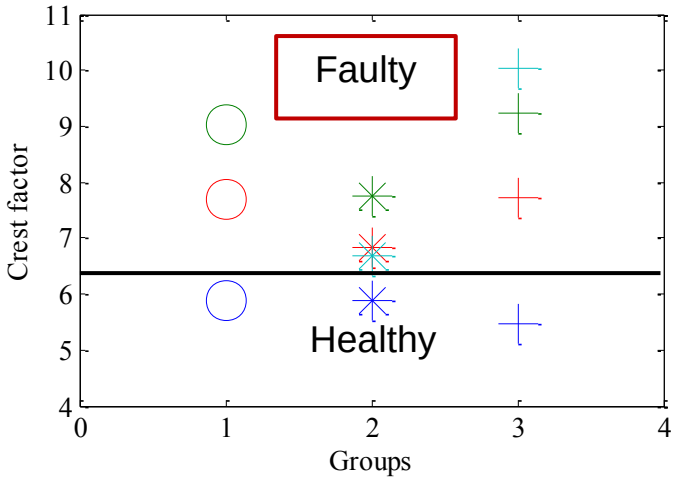
4.4.3 Conventional time domain indicators

To more accurately diagnose rope fault, we further evaluate the effectiveness of 11 common conventional time domain indicators in detecting aforementioned three rope faults. The results are listed in Table 4.1.

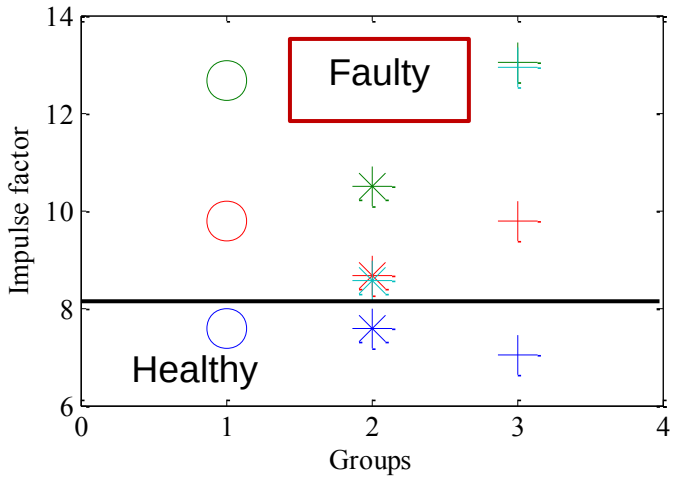
Table 4.1 Evaluation of time domain indicators

Indicators	Healthy	Fatigue	Bump	Breakage
Max	0.155	0.322	0.260	0.288
Mean	0.022	0.024	0.026	0.022
Rms	0.028	0.034	0.033	0.028
Skewness	7.196	1.256	5.524	6.715
Kurtosis	2.426	1.160	4.784	2.488
Shape factor	1.287	1.407	1.269	1.286
Crest factor	5.473	9.250	7.707	10.042
Impulse factor	7.044	13.021	9.781	12.922
Clearance factor	8.433	16.183	11.556	15.460
Kurtosis factor	3.676	7.899	3.682	3.647
Vibration severity	309.629	291.711	313.410	307.542

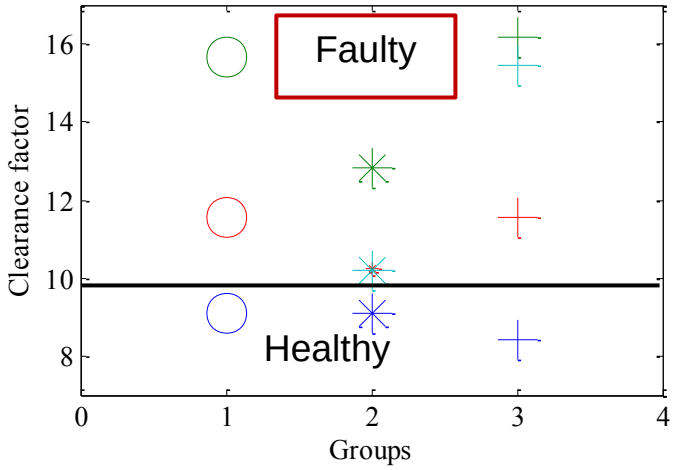
According to the value of these 11 indicators under 4 different health conditions, we found 3 indicators (as labeled in red rectangle) capable of classifying the health condition, including crest factor, impulse factor and clearance factor. The usage of these time domain indicators in classifying fault is illustrated in Figure 4.15.



(a) Crest factor



(b) Impulse factor



(c) Clearance factor

Fig 4.15 Usage of time domain indicators

Group 1-3 which serve as x label in Figure 4.15 stand for different data groups in analysis. In each group (marked with same shape), 4 signals representing healthy (blue one) and other 3 faulty cases are analyzed. Three indicators (crest, impulse and clearance factor) are studied in classifying healthy and faulty conditions, using 3 groups of data. The black line is a coarse dividing line. According to the distribution of indicator value, these three indicators (crest factor, impulse factor and clearance factor) are effective in classifying these three rope faults (rope fatigue, rope local bump and rope local breakage).

4.5 Compound fault diagnosis methods

4.5.1 Compound indicator K_1 and K_2

Based on former validated 5 most effective time domain as well as frequency domain indicators (see Figure 4.16), we try to propose a more simple and efficient indicator for real world elevator fault diagnosis.

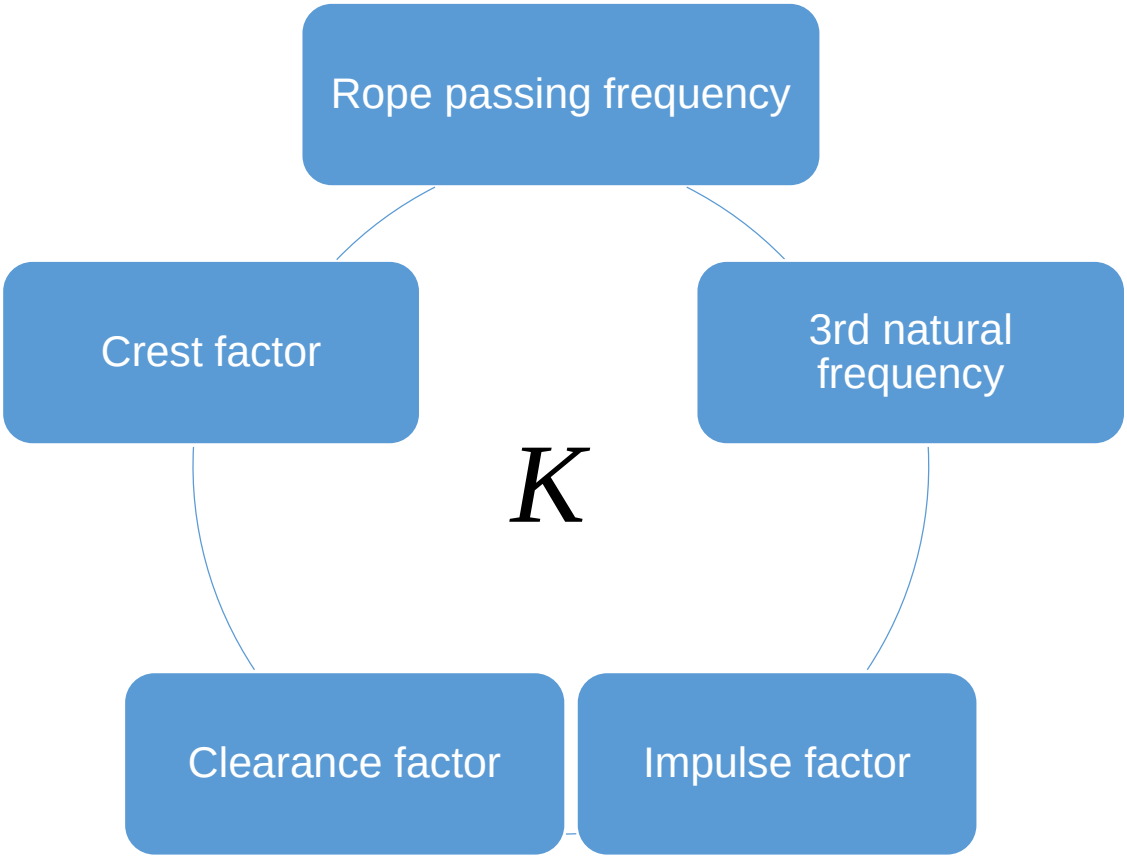


Fig 4.16 Usage of time domain indicators

(1) ***K*₁*:***

By integrating abovementioned 6 time domain and frequency domain indicators, we propose an integrated indicator defined by equation 4.7.

$$\begin{aligned} K_1 &= \frac{k_a}{k_a^0} \cdot \frac{k_b}{k_b^0} \cdot \frac{k_c}{k_c^0} \cdot \frac{k_p}{k_p^0} \cdot \frac{k_n^0}{k_n} \\ &= \prod_{i=1}^{5} \frac{k_i}{k_i^0} \cdot \frac{k_n^0}{k_n} \end{aligned}, \tag{4.7}$$

where k_a k_b k_c stands for the three time domain indicators, and k_p k_n respectively means the rope passing frequency and natural frequency. Symbol labeled with an extra superscript 0 is the corresponding rated value. Its properties are summarized as:

(1) On ideal healthy condition, k=1;
(2) When fault exits, the value of this indicator increase;
(3) The value of k can be used to judge the fault severity.

(2) ***K*₂*:***

Another integrated indicator is defined by equation 4.8.

$$\begin{aligned} K_2 &= \frac{\left|k_a - k_a^0\right|}{k_a^0} + \frac{\left|k_b - k_b^0\right|}{k_b^0} + \frac{\left|k_c - k_c^0\right|}{k_c^0} + \\ &\quad \frac{\left|k_p - k_p^0\right|}{k_p^0} + \frac{\left|k_n - k_n^0\right|}{k_n^0} \\ &= \sum_{i=1}^{6} \frac{\left|k_i - k_i^0\right|}{k_i^0} \end{aligned}, \tag{4.8}$$

Where k_a k_b k_c stands for the three time domain indicators, and k_p k_n respectively means the rope passing frequency and natural frequency. Symbol labeled with an extra superscript 0 is the corresponding rated value. Its properties are summarized as:

(1) On ideal healthy condition, k=0;
(2) When fault exits, the value of this indicator increase;
(3) The value of k can be used to judge the fault severity.

4.5.2 Energy ratio histogram

A new histogram indicator termed energy ratio histogram is proposed to integrate the three feature frequency indicator as an intuitive and practical monitoring tool, as shown in Figure 4.17.

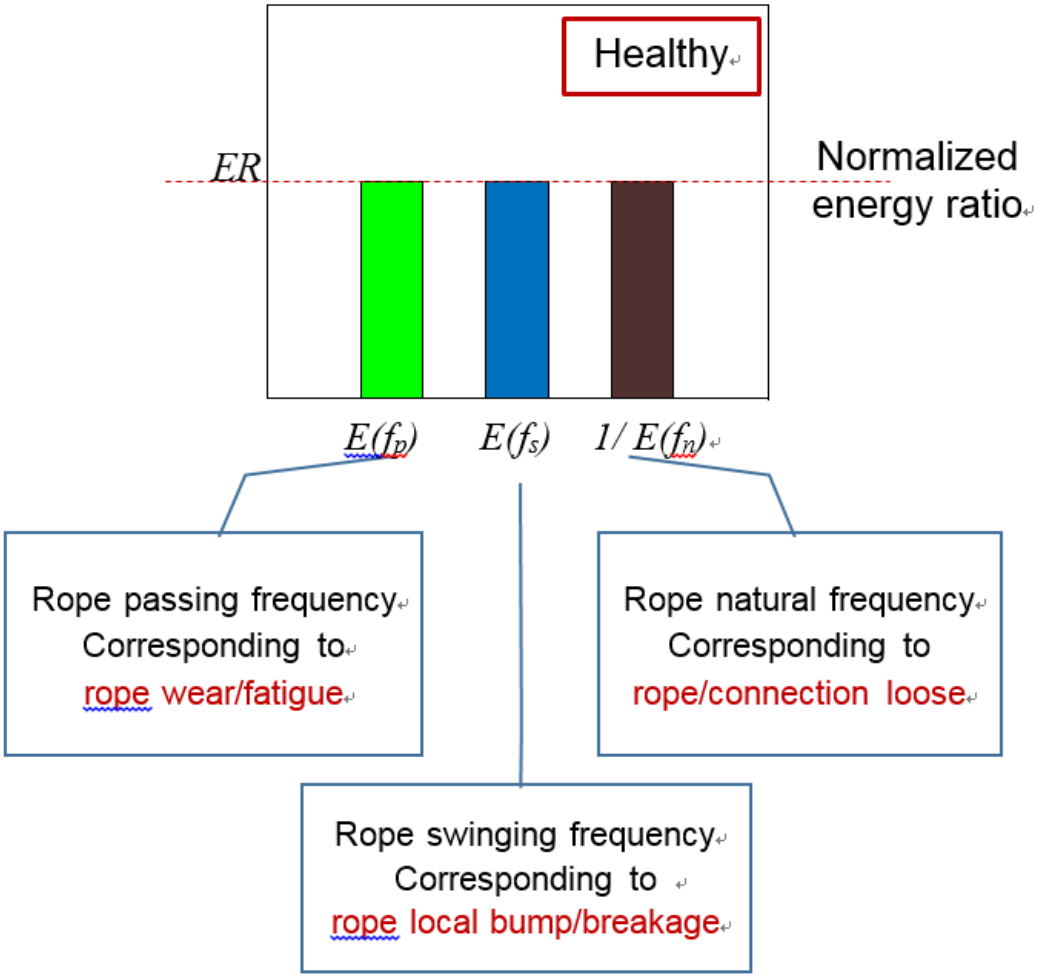


Fig 4.17 Energy ratio histogram

Three bar standing for aforementioned two feature frequency indicators and a rope swing frequency (will be further studied in next phase) compose the energy ratio histogram. Actually energy ratio is a normalization method to set these three indicators to value 1. Since rope passing frequency is specific for distributed rope fault diagnosis like rope wear, rope swinging frequency is specific for local rope fault diagnosis like rope local breakage, and rope natural frequency is specific for rope connection fault diagnosis like rope loose, this histogram integrates these three main categories of fault diagnosis methods and provide a simple yet thorough method for monitoring rope condition and diagnose the fault type. As shown in Figure 4.18, each bar stands for a type of rope fault severity, which is quite easy for maintainer.

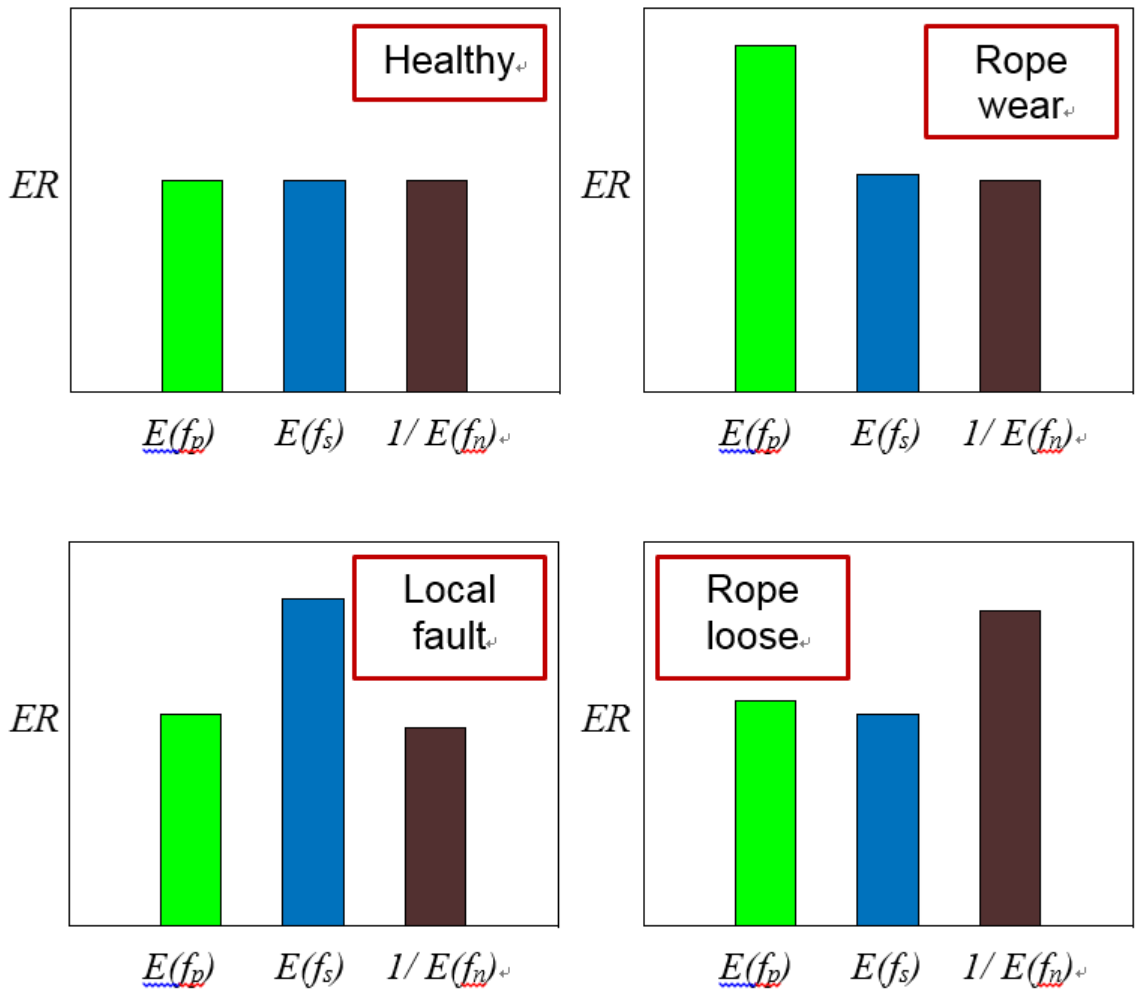
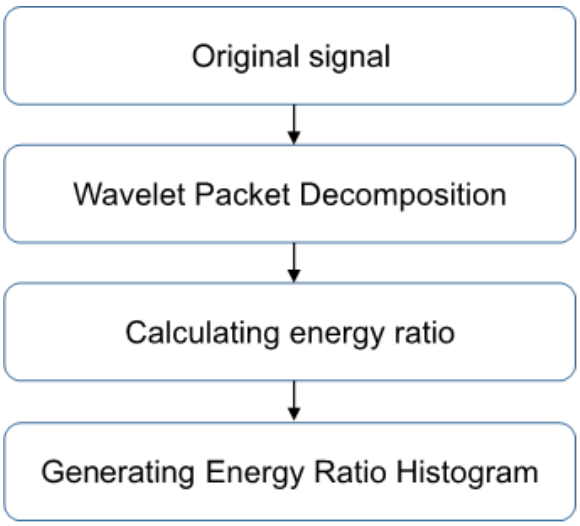


Fig 4.18 Usage of energy ratio histogram

The procedure of using energy ratio histogram to monitor rope condition is summarized as:



4.6 Summary

4.6.1 Summary for time domain indicators

In this section we have studied the effectiveness of conventional time domain analysis and indicators in diagnosing 3 types of rope fault by analyzing rope tester signals. Conclusions can be drawn as

- Pulley signal waveform features prominent impulses in case of rope fault condition. However, waveform detection is not capable of diagnosing weak or early-stage fault.
- Probability density distribution is probably the simplest fault distinguish method, but it is not fit for precise diagnosis.
- Crest factor, impulse factor and clearance factor are three most effective indicators among 11 common indicators, but any one of them cannot be directly used without life test and improvement.
- Conventional indicators is useful in elevator system fault diagnosis especially rope diagnosis, but new indicator based on elevator properties is suggested in real world application.

4.6.2 Summary for frequency domain indicators

In this section we have proposed 2 new frequency domain indicators which are found during our former data analysis process. The definition, property, validation and procedure of using these 2 feature frequency indicators have been illustrated. Conclusions can be drawn as

- Rope passing frequency is an important characteristic frequency result from rope-pulley contact. It can be observed in Fourier spectrum of both motor and pulley (which have sheave) vibration signals.
- By monitoring the amplitude of rope passing frequency, and integrating the rope life-time process, we can evaluate the health condition of rope.
- Techniques to enhance the obviousness of rope passing frequency is also proposed as a reference.
- System natural frequency is sensitive to stiffness change, therefore can be used to diagnose rope tension lose or rope deformation. Besides, frequency range energy E(f1,f2) and energy ratio ER(f1,f2) are proposed as improved version.

4.6.3 Notes

- All above proposed indicators are based on either rope tester signals, pilot elevator signals, or numerical simulated elevator model.
- However, the threshold of these indicators in real world application still demands large amount of experiment and comparison between different fault severity cases.
- All indicators should be further validated in experiment as well as real application, and then adjust the parameters for better effectiveness.

5 MOTOR FAULT DIAGNOSIS

Traction motor is the power source and its sheave connects to the car with hoist rope, thus fault induced abnormal vibration in motor will transmit to the car, harming the car smoothness. Hence, the motor condition monitoring and fault diagnosis is also an important topic.

5.1 Motor bearing fault diagnosis

5.1.1 Pilot elevator motor

In pilot elevator vibration test, we measure the traction motor vibration with accelerometers mounted on motor casing to study the characteristics of motor in an elevator system. The measuring point is shown in Figure 5.1, and information of motor bearing is:

Diameter: 480mm
Current frequency: 21.2Hz
Rotating speed: 127.3Rpm, i.e., 2.12Hz
Number of pole pairs: 10
Bearing type: SKF6218-2RS1

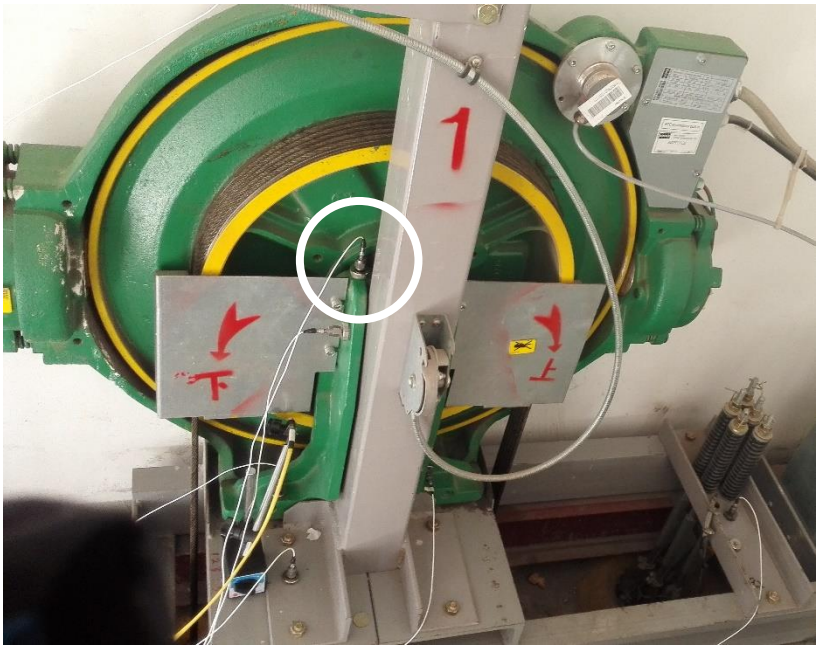


Fig 5.1 Motor measuring point for pilot elevator test

The motor bearing fault frequencies can be calculated based on its size and rotating speed information, as:

$$f_{\mathrm{i}} = \frac{1}{2} Z(1+\frac{d}{D}\cos\alpha) f_r = \frac{1}{2} * 10 * (1+\frac{35}{125}\cos 0) * 2.12 = 13.7\,\mathrm{Hz}\,, \tag{5.1}$$

$$f_{\mathrm{o}} = \frac{1}{2} Z(1-\frac{d}{D}\cos\alpha) f_r = \frac{1}{2} * 10 * (1-\frac{35}{125}\cos 0) * 2.12 = 9.67\,\mathrm{Hz}\,, \tag{5.2}$$

$$f_b = \frac{1}{2}\frac{D}{d}(1-\frac{d}{D}\cos\alpha)^2 f_r = \frac{1}{2} * \frac{125}{35} * (1-\frac{35}{125}\cos 0)^2 * 2.12 = 1.96\,\mathrm{Hz}\,, \tag{5.3}$$

$$f_{\mathrm{retainer}} = \frac{1}{2}(1+\frac{d}{D}\cos\alpha) f_r = \frac{1}{2} * (1+\frac{35}{125}\cos 0) * 2.12 = 0.29\,\mathrm{Hz}\,, \tag{5.4}$$

where f_{i}, f_{o}, f_{b} and f_{retainer} respectively stand for the bearing inner ring fault frequency, outer ring fault frequency, running element fault frequency and retainer fault frequency. These calculations are used for frequency component identification.

The envelope spectrum has merit of simple frequency structure, thus is widely used in fault diagnosis field. The envelope spectrum of motor vibration signal is plotted in Figure 5.2, from which we can only observe two dominant frequency components and their combinations. They are further validated as two times the current frequency and an 8Hz unknown component related to electricity. However, the bearing information is still concealed due to strong interference by electricity.

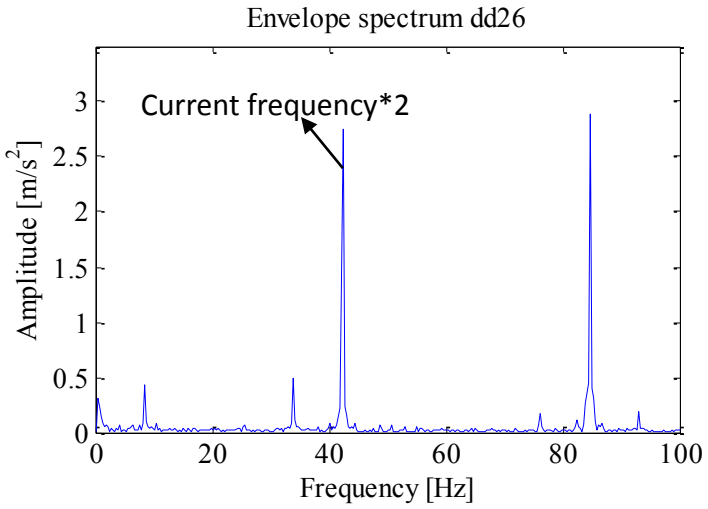


Fig 5.2 Envelope spectrum of motor vibrations signal

To eliminate the interference by electricity, we try to find out the electricity–related frequency components. In the Fourier spectrum of motor vibration signal as shown in Figure 5.3 we find some high frequency components locate around 5500Hz, whose frequency intervals correspond to the current frequency. Thus these high frequency components result from electricity factors and can be filtered out.

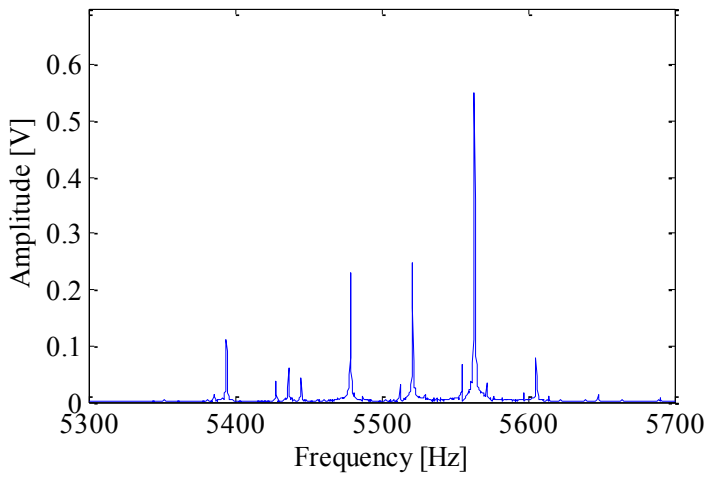


Fig 5.3 Fourier spectrum of motor vibrations signal (around 5500Hz)

The original motor vibration signal is filtered with a low–pass filter and its envelope spectrum is plotted in Figure 5.4. Clearly more detailed frequency component information are observed in envelope spectrum of filtered motor vibration signal. Besides, the bearing inner ring fault frequency and its double are pinpointed, which implies the inner ring fault.

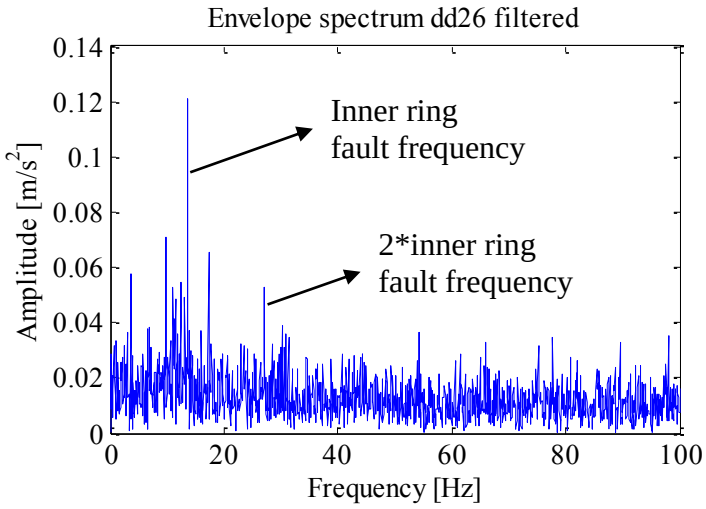


Fig 5.4 Envelope spectrum of filtered motor vibrations signal

5.1.2 Lab motor

Further we carried out vibration test on an independent motor in Kunshan lab to eliminate the influence brought in by elevator system. The measuring point is shown in Figure 5.5, and information of this motor bearing is:

Diameter: 420mm
Current frequency: 75.8Hz
Rotating speed: 181.9Rpm, i.e., 3.03Hz
Number of pole pairs: 25
Bearing type: SKF6220-2RS1

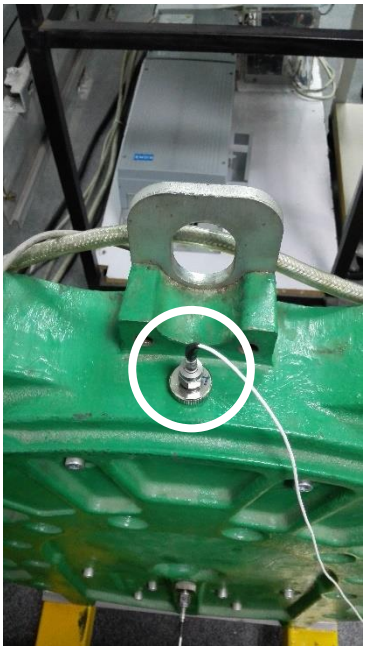


Fig 5.5 Motor measuring point for lab experimental test

Similarly, the motor bearing fault frequencies can be calculated based on its size and rotating speed information, as:

$$f_{\mathrm{i}} = \frac{1}{2} Z(1+\frac{d}{D}\cos\alpha) f_r = \frac{1}{2} * 10 * (1+\frac{21}{139}\cos 0) * 3.03 = 17.43\,\mathrm{Hz}\,, \tag{5.5}$$

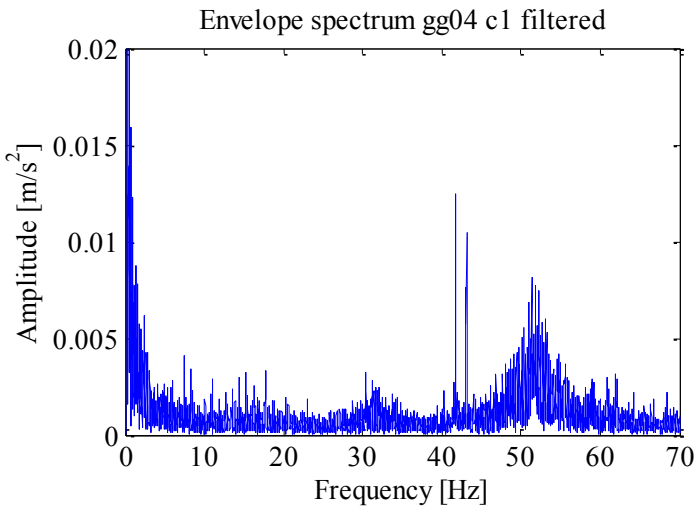
$$f_{\mathrm{o}} = \frac{1}{2} Z(1-\frac{d}{D}\cos\alpha) f_r = \frac{1}{2} * 10 * (1-\frac{21}{139}\cos 0) * 3.03 = 12.86\,\mathrm{Hz}\,, \tag{5.6}$$

$$f_b = \frac{1}{2}\frac{D}{d}(1-\frac{d}{D}\cos\alpha)^2 f_r = \frac{1}{2} * \frac{139}{21} * (1-\frac{21}{139}\cos 0)^2 * 3.03 = 7.22\,\mathrm{Hz}\,, \tag{5.7}$$

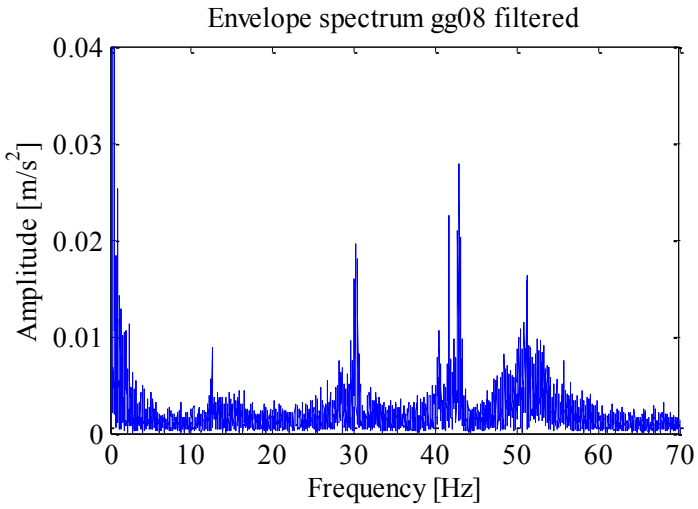
$$f_{\mathrm{retainer}} = \frac{1}{2}(1+\frac{d}{D}\cos\alpha) f_r = \frac{1}{2} * (1+\frac{21}{139}\cos 0) * 3.03 = 1.74\,\mathrm{Hz}\,, \tag{5.8}$$

where f_{i}, f_{o}, f_{b} and f_{retainer} respectively stand for the bearing inner ring fault frequency, outer ring fault frequency, running element fault frequency and retainer fault frequency. These calculations are used for frequency component identification.

Again we perform a low–pass filtering before envelope spectrum analysis. The envelope spectrum of filtered motor vibration signal under two different running speed conditions are exhibited in Figure 5.6 (a) and (b). However, we find that the dominant frequency components in both low speed and high speed conditions are same in frequency value and have different amplitude. It indicates that these dominant frequency components are not correlated to bearing and are most likely the natural frequencies. Modal test on independent motor further validates the identification. Hence, this motor bearing is probably in its health condition.



(a) Low speed condition

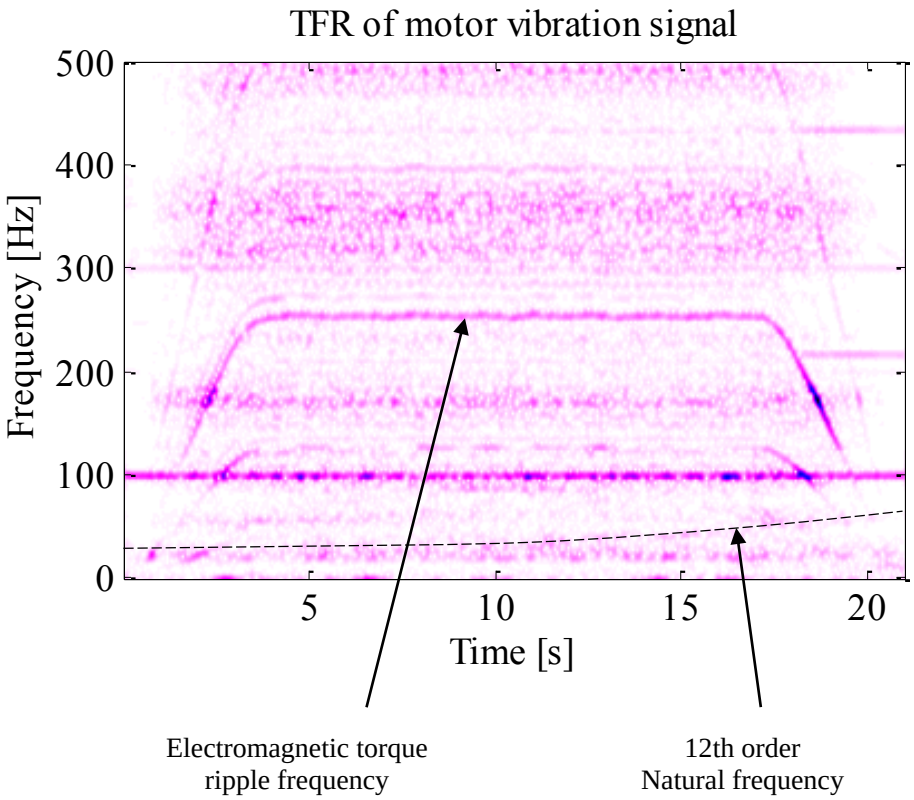


(b) high speed condition

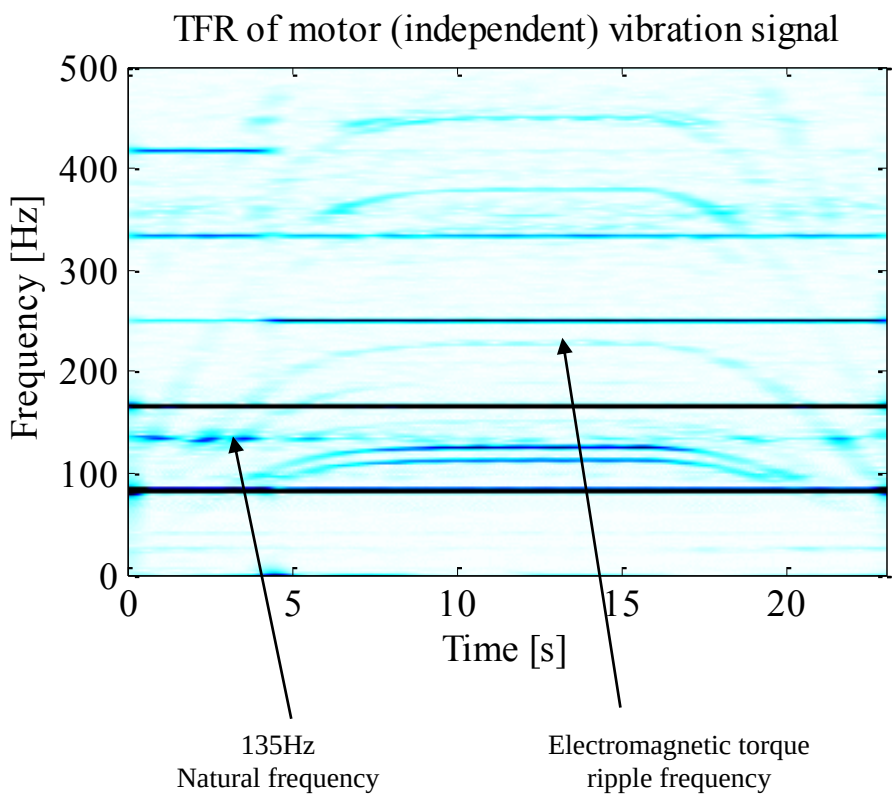
Fig 5.6 Envelope spectrum of filtered motor vibration signal

5.2 Frequency structure of motor vibration signal

In this section we employ the joint time–frequency domain analysis method to give insight to the frequency structure of motor vibration signal. The short–time Fourier transform (STFT) is applied to both motor vibration signals in pilot elevator test and lab experimental test, and their time–frequency representations (TFRs) are presented in Figure 5.7 (a) and (b).



(a) pilot elevator test



(b) lab experimental test

Fig 5.7 TFRs of motor vibration signal

TFR reveals the frequency component in both time and frequency domain, thus is suitable for analyzing nonstationary signals. From the TFR in Figure 5.7 we can clearly identify the main frequency components in motor vibration signals. Except for those inherent frequency components which correlate to electricity, we can verify two types of characteristic frequency components, including system natural frequency and electromagnetic ripple frequency. Based on a brief understanding of the frequency structure of motor vibration signal, we can use the TFR for fault feature frequency components extraction.

5.3 Summary

Based on analysis of vibration signals collected from both pilot elevator motor and lab independent motor, we can conclude that:

- Due to the electricity interference, we propose a method integrating low–pass filtering and envelope spectrum to extract motor bearing fault frequency components. It can be used to diagnose motor bearing fault, and by evaluating the amplitude of main fault frequency components we can also monitor bearing

condition in real time.

- The frequency structure of motor vibration is complex, while system natural frequency and electromagnetic ripple frequency are dominant and useful in system modal analysis as well as running speed estimation.

6 GUIDE RAIL/SHOE FAULT DIAGNOSIS

Improving elevator ride quality in horizontal direction is the urgent need for guaranteeing passenger comfort. Since the contact between guide rail and guide shoe is the main external factor which result in abnormal horizontal vibration/shock, in this section we studied the specific fault diagnosis method for guide rail/shoe.

6.1 Data acquisition

Horizontal vibration analysis is an important part of elevator vibration analysis. Since the guide rail/shoe is the direct contact part and guide rail is fixed and only contact with guide shoe, **we propose to collect and analyze the guide rail horizontal vibration signal which is clear and reflect much information of guide rail/shoe.** We measure the guide rail horizontal vibration signal in the pilot elevator test with 4 wireless accelerometers evenly distributed on the guide rail, as shown in Figure 6.1.

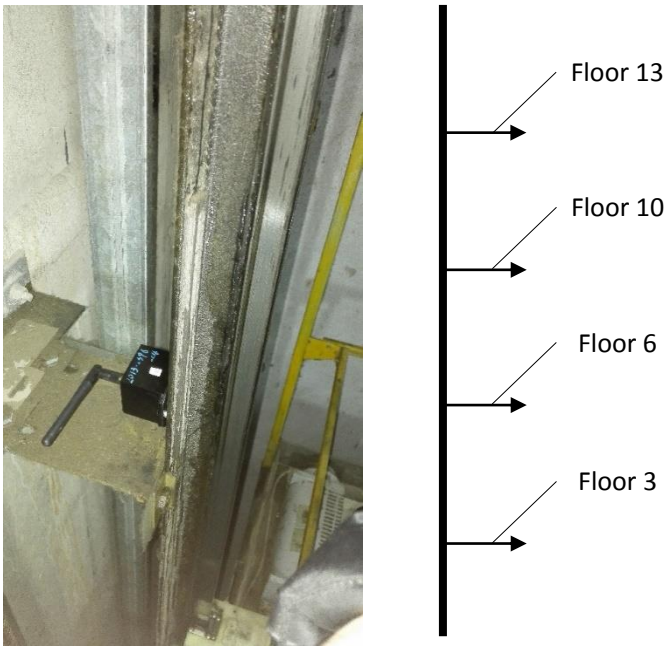
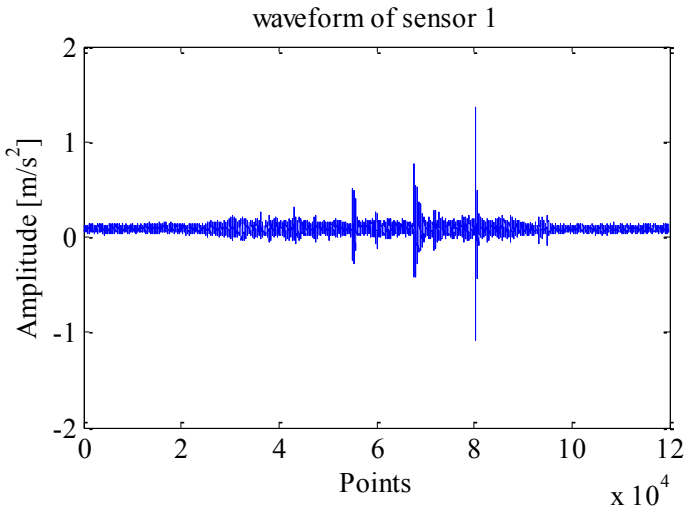


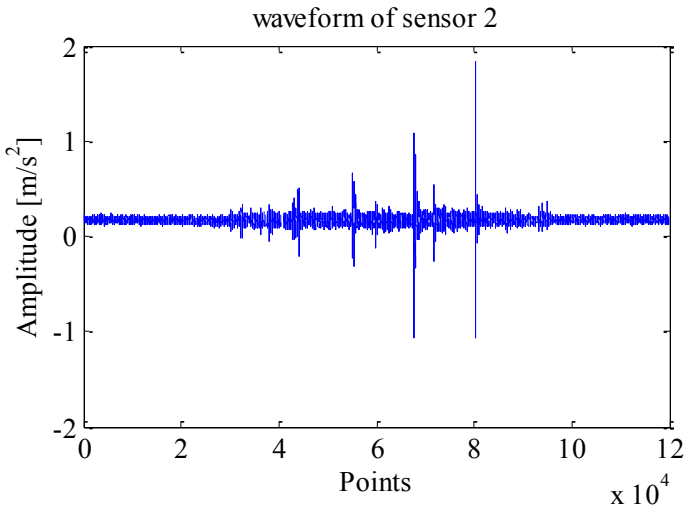
Fig 6.1 Guide rail measuring point for pilot elevator test

6.2 Guide rail vibration signal

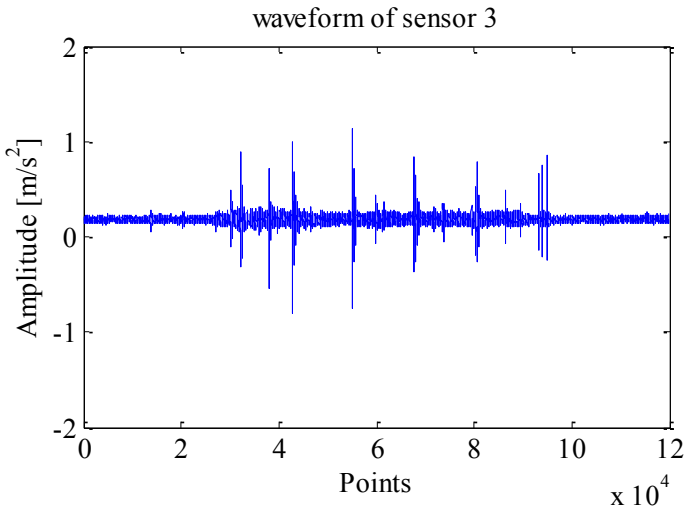
Guide rail vibration signals have weak periodicity since horizontal vibration is weak in ideal situation. However, the evident shock are expected once fault exist in guide rail or guide shoe. The waveform of guide rail vibration signal contains abundant of guide rail health information, as plotted in Figure 6.2.



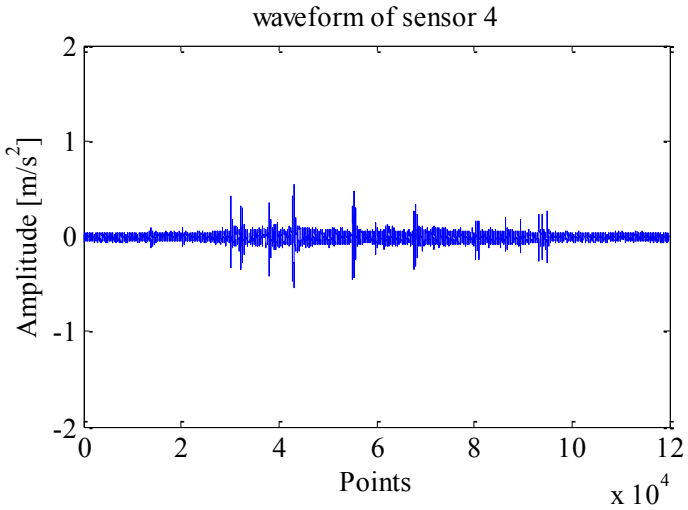
(a) Sensor 1



(b) Sensor 2



(c) Sensor 3



(d) Sensor 4

Fig 6.2 Vertical free vibration

In Figure 6.2(a) plots the waveform of guide rail vibration signals collected by 4 accelerometers mounted on guide rail at different height. Clearly we observe some big impulses which have similar time interval. Actually they are caused by the weld joint between adjacent guide rail parts. Since the height interval between adjacent weld joint is fixed, the time interval between adjacent impulses is only decided by car speed. Except for the inherent impulses caused by weld joint, we focus on the abnormal impulses which reflect guide rail fault. The detail of guide rail/shoe fault diagnosis method will be introduced in the following.

6.3 Guide rail/shoe fault diagnosis

For horizontal vibration monitoring and fault diagnosis, time domain analysis is more efficient since fault mainly result in horizontal random impulse. Both an time

domain indicator for preliminary fault identification and statistic impulse recording for fault location are proposed.

6.3.1 Fault detection

The kurtosis index K is commonly used for abnormal impulse detection. When more impulses arise K increase correspondingly. The definition of kurtosis index is derived as

$$K_v = \frac{\beta}{X_{rms}^4}, \tag{6.1}$$

$$\beta = \frac{1}{N}\sum_{i=1}^{N} x_i^4, \tag{6.2}$$

$$X_{rms} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}, \tag{6.3}$$

Where K is the kurtosis index, β stands for the kurtosis and X_{rms} means the root mean square.

When local damage exists on guide rail or guide shoe, or the connection is over tight or over loose, more impulses will occur and corresponding kurtosis value will become larger. Setting a baseline value and upper limit, we can obtain and record the everyday kurtosis index for condition evaluation and fault prediction.

Numerical simulation was performed to validate the effectiveness of kurtosis index in detecting weak fault. As illustrated in Figure 6.3, we use the sensor 3 in–situ vibration signal as the baseline and add 8 different kinds of impulses manually to examine the change of kurtosis. The red lines pinpoint the time instant for extra pulses. Since the original signal impulse caused by weld joint is around 1 in amplitude, we introduce 4 mono impulses with different amplitude of 2, 1, 0.6 and 0.4. In addition, we introduce a condition that 3 impulses exist concurrently. Thus 8 impulse conditions in total are simulated, and the corresponding result of kurtosis value, kurtosis and average are listed in Table 6.1.

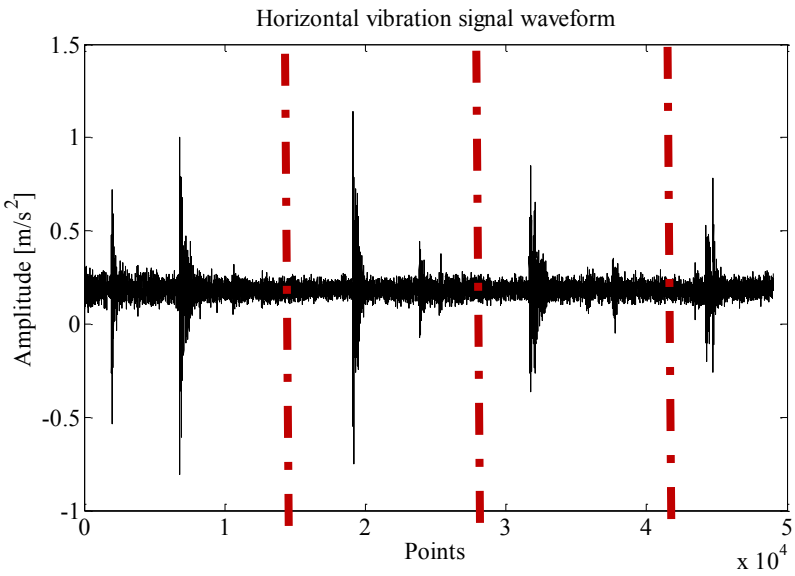


Fig 6.3 Illustration of impulse simulation

Table 6.1 Result of impulse simulation

Extra impulse amplitude	Kurtosis	Kurtosis value	Average
Baseline	0.0022	1.6053	0.1844
2	0.0025	1.8344	0.1844
1	0.0022	1.6184	0.1844
0.6	0.0022	1.6066	0.1844
0.4	0.0022	1.6054	0.1844
2 (3 impulses)	0.0032	2.2867	0.1845
1 (3 impulses)	0.0023	1.6445	0.1844
0.6 (3 impulses)	0.0022	1.6093	0.1844
0.4 (3 impulses)	0.0022	1.6057	0.1844

From the simulation result listed in Table 6.1 we find that when weak impulse arises, the kurtosis and average value of the signal almost remain the same, while the kurtosis value increases once extra impulse is brought in. Besides, when more or stronger impulses are added, the kurtosis increases more. It is a good quality for fault detection indicators. Hence, we monitor the kurtosis index to detect guide rail fault. The usage of this indicator is exhibited in Figure 6.4 and the procedure is also listed.

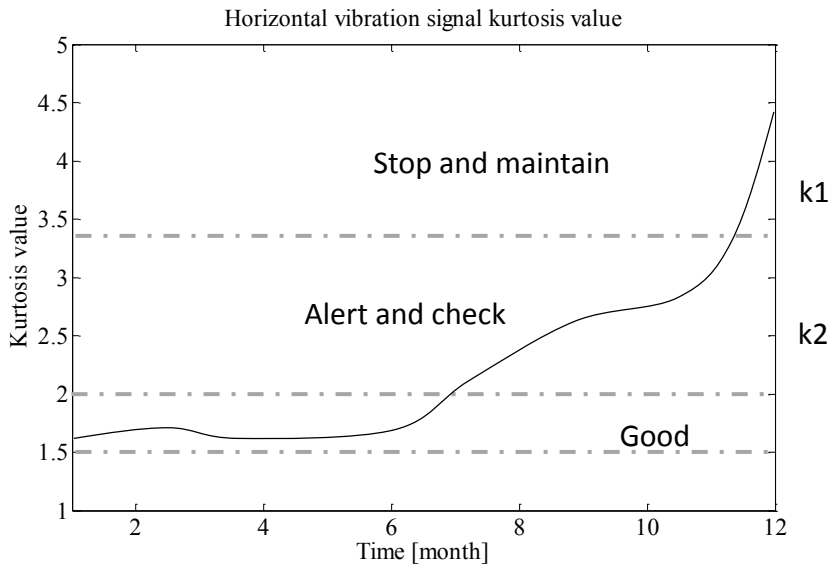


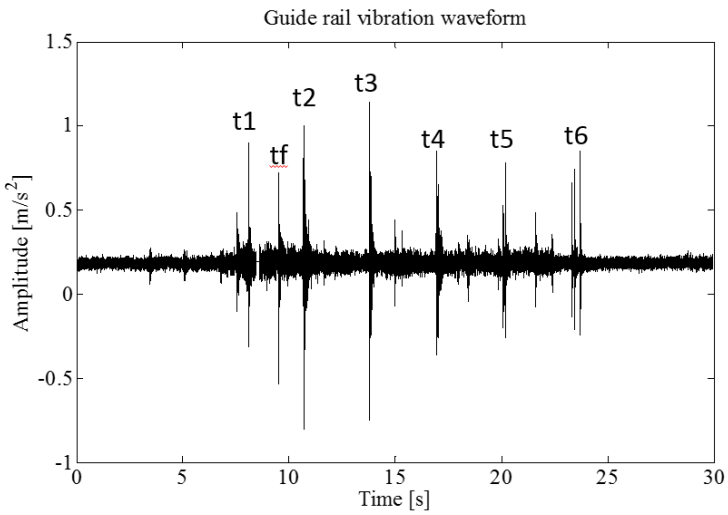
Fig 6.4 Usage of kurtosis index

The procedure is summarized as:

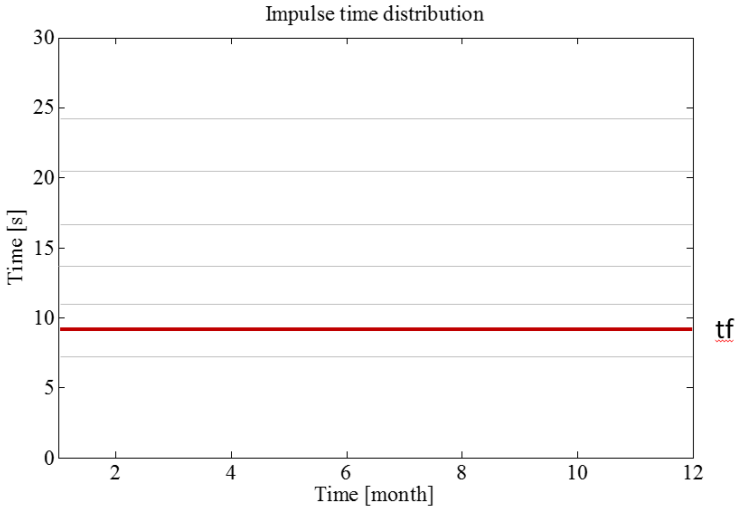
(1) Mount an accelerometer on the guide rail to collect guide rail horizontal vibration signal;
(2) Obtain the value of kurtosis index of the collected signal by calculation;
(3) Record the value of indicator every constant times and get the monitoring chart;
(4) Take moves when the indicator has obvious varying trend.

6.3.2 Fault localization

In order to accurately locate the local fault, another work is to recording the time instant of impulses during the car travel. Note that the impulses result from guide rail joint is inevitable but can be accurately predicted, these time instant can be ignored. The usage of recording impulse time instant for local fault localization is illustrated in Figure 6.5.



(a) Waveform



(b) Impulse time instant distribution

Fig 6.5 Usage of impulse time instant

In Figure 6.5(a) we can observe a series of impulses in waveform. Actually some high impulses like t_1–t_6 correspond to weld joint influence, while abnormal impulses like t_f indicate local fault. By recording the time instant t_f, we can obtain the vibration source of the impulse according to the velocity of car. Thus the location of the fault is achieved. The procedure of using time recording method for horizontal local fault identification is given as

(1) Set the standard for self checking travel;
(2) Set a threshold for computer to search for impulses automatically and record the time vector t;
(3) Record the time instant of impulses during self checking every day or month, and plot the time distribution of impulses;

(4) Check the time distribution chart, if an impulse occurs at the same time every day, i.e., a clear line exist on the distribution chart, a fault is extracted. The location can be calculated with car speed information.

6.4 Summary

Horizontal fault is mainly manifested as impulses which can be detected and exploited in time domain analysis. Two methods including dimensionless statistic indicator (kurtosis value) and impulse time tracking are proposed to identify horizontal fault existence as well as accurately locate the fault.

Based on guide rail vibration signal measured in pilot elevator test and the numerical simulation, we can conclude that:

- Guide rail vibration signal features evident impulses result from weld joint. The time interval of these impulses are decided by car speed.
- Kurtosis index is validated to be effective in detecting abnormal impulses. By recording the value of kurtosis index we can monitor the health condition of guide rail/shoe structure.
- An impulse time instant recording method is proposed to further localize the guide rail local fault. With the time instant of abnormal impulse and corresponding car speed, we can calculate the position of guide rail local fault.

7 APPLICATION SCHEME

In Section 3–6 the vibration–based fault diagnosis of hoist rope, traction motor and guide rail/shoe have been studied and specific diagnosis methods as well as indicators have been proposed. In order to integrate all these methods and indicators in general and convenient applications, in this section we propose a general self–checking scenario and a fault database scenario for elevator system condition monitoring and fault diagnosis.

7.1 Self–checking scenario

7.1.1 Running Conditions

The merit of using self–checking process to perform all aforementioned diagnosis method is that all the obtained indicators are based on same working condition and thus own comparability. Taking the configuration parameters into consideration, the running condition for the pilot elevator is proposed as

Principle: Safety and without interruption
Time: around 3:00 a.m. every day to ensure safety
Condition: empty load, fixed maximum speed
Sensors: 5 accelerometers, refer to vibration measurement method in Section 2

7.1.2 Self–checking scenario

A series of movement are required in a self–checking process to collect abundant elevator system information. Two sets of self–checking scenario are exhibited in Table 7.1 and Table 7.2, while the former employs modal test for natural frequency identification and the latter extract natural frequency information from travel test.

Table 7.1 Self checking scenario (draft, both modal test and travel test)

Travel	Meaning	Indicator	Sensor location	Sensor type	Use
Floor-15	Preparation				
15-14(13)	Modal test	Frequency peak f1	Car frame (vertical)	Accelerometer	Rope condition monitoring
14-13	Adjustment				
13-3	Travel test	Bearing frequency amplitude a1-a4	Motor (vertical)	Accelerometer	Motor bearing condition monitoring
		Passing frequency amplitude a5	Motor (vertical)	Accelerometer	Rope condition monitoring (overall wear)
		Bearing frequency amplitude a6-a9	Pulley (axial)	Accelerometer	Pulley bearing condition monitoring
		Passing frequency amplitude a10	Pulley (axial)	Accelerometer	Rope condition monitoring (overall wear)
		Impulse time t1	Pulley (axial)	Accelerometer	Rope/pulley condition monitoring (local damage)
		Kurtosis value V1	Pulley (axial)	Accelerometer	Rope condition monitoring (rope wear)
		Kurtosis value V2	Guide rail	Accelerometer	Guide rail condition monitoring
		Impulse time t2	Guide rail	Accelerometer	Guide rail condition monitoring (local damage)
3-2(1)	Modal test	Frequency peak f2	Car frame (vertical)	Accelerometer	Rope condition monitoring
2-1	Adjustment				
1-15	Safety check				
15-1					
Calculate integrated indicator K1 and K2, draw the energy ratio histogram, check the result for overall condition evaluation.					

Table 7.2 Self checking scenario (draft, travel test only)

Travel	Meaning	Indicator	Sensor location	Sensor type	Use
Floor-15	Preparation				
13-3	Travel test	Frequency peak f1	Car frame (vertical)	Accelerometer	Rope condition monitoring
		Bearing frequency amplitude a1-a4	Motor (vertical)	Accelerometer	Motor bearing condition monitoring
		Passing frequency amplitude a5	Motor (vertical)	Accelerometer	Rope condition monitoring (rope wear)
		Bearing frequency amplitude a6-a9	Pulley (axial)	Accelerometer	Pulley bearing condition monitoring
		Passing frequency amplitude a10	Pulley (axial)	Accelerometer	Rope condition monitoring (rope wear)
		Impulse time t1	Pulley (axial)	Accelerometer	Rope/pulley condition monitoring (local damage)
		Kurtosis value V1	Pulley (axial)	Accelerometer	Rope condition monitoring (rope wear)
		Kurtosis value V2	Guide rail	Accelerometer	Guide rail condition monitoring
		Impulse time t2	Guide rail	Accelerometer	Guide rail condition monitoring (local damage)
		Frequency peak f2	Car frame (vertical)	Accelerometer	Rope condition monitoring
3-floor	Restoration				
Calculate integrated indicator K1 and K2, draw the energy ratio histogram, check the result for overall condition evaluation.					

7.2 Fault database

The self–checking scenario integrates all proposed method into a series of test process. After the measurement and calculation of all aforementioned indicators, we still need a simple and legible document for elevator maintainer to understand the real health condition of the whole elevator system. The meaning of each indicator, corresponding fault information and recommend solution are integrated in the fault database, see Table 7.3.

Table 7.3 Self checking scenario (draft, travel test only)

Fault location	Indicator	Symptom	Result	Solution
Traction motor	a1-a4	Exceed upper limit	Bearing fault	Change bearing
Pulley	a6-a9	Exceed upper limit	Bearing fault	Change bearing
	t1	Repeat often	Pulley/rope local fault	Calculate location and check
Rope	K1	monotone increasing	Rope fault	Check rope
	K2	monotone increasing		
	Histogram,	Abnormal in height		
	a5, a10	Exceed lower/upper limit	Rope wear	Check rope
	f1, f2	Exceed lower limit	Rope loose/wear	Refer to f1-f2 and check rope
	f1-f2	monotone increasing	Car side rope local fault	Check car side rope
	f1-f2	monotone decreasing	Counterweight side rope local fault	Check counterweight side rope
	V1	monotone increasing	Rope wear	Check rope
Guide rail/shoe	V2	Exceed upper limit	Guide rail local fault	Refer to t2 and check
	t2	Repeat often	Guide rail local fault	Calculate location and check
Unfinished				

8 CONCLUSION AND FUTURE WORK

- To evaluate elevator health condition and diagnose potential fault based on vibration analysis, we have carried out test on both pilot elevator and lab rope tester to measure vibration signals from traction motor, car, car frame, counterweight, pulley and guide rail.
- According to measured datasets, we have employed time domain, frequency domain and joint time–frequency domain analysis to study fault characteristics.
- Focus on rope condition, we evaluate the effectiveness of 11 conventional time domain indicators and extract 3 most effective ones for fault detection.
- Three feature frequency indicators, i.e., rope passing frequency and 3rd order natural frequency have been discovered and successfully applied in diagnosis of 2 types of rope fault.
- Integrating both time domain and frequency domain indicators, we have proposed two compound indicators and 1 energy ratio histogram for convenient fault diagnosis.
- For motor bearing fault diagnosis, we have applied low–pass filtering and envelope sepctrum to detect bearing inner ring fault.
- For guide rail/shoe fault diagnosis, we have proposed a kurtosis index indicator for fault detection and a impulse time recording method for fault localization.
- Combing all diagnosis methods and indicators for main elevator components, we have proposed a self–checking scenario and a fault database for real world application.

The research and application work in phase 2 have fulfilled the expectation in phase 1 report. To further apply all these proposed methods and indicators for elevator condition monitoring and fault diagnosis, we look forward to carrying out more tests on real elevator or elevator test rig, especially life test and real world fault simulation. The core content for phase 3 is still the improvement and summary of condition monitoring and fault diagnosis for main component of the elevator, with both theory and application proposals.

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